Synergistic Policy Intervention for Green Investment and Bond Pricing: An Optimization Framework with Carbon Credits and Subsidies

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Abstract

Green projects often generate environmental benefits that are not fully reflected in private cash flows, creating a wedge between social value and financial viability. This paper develops a two-stage framework that links policy intervention to green bond pricing. In the first stage, we formulate a constrained policy-optimization model that allocates carbon credits and fiscal subsidies to restore project viability under carbon-credit supply constraints, subsidy budget constraints, and institutional limits. The optimization balances discounted policy-preference cost with nominal lifecycle policy resource commitment, thereby embedding a market-first hierarchy in which carbon-credit revenues are prioritized and subsidies are used as a fiscal backstop. In the second stage, we map the optimized policy-adjusted cash flows into an option-pricing model to value the embedded green attribute of green bonds. The framework shows how policy support repairs downside cash-flow shortfalls, shifts green assets from a distressed out-of-the-money state toward an at-the-money state, and allows uncertainty in future environmental value to be capitalized as upside optionality. The empirical results show that carbon credits and subsidies are complementary rather than interchangeable. Carbon credits can restore viability when verified emission reductions are sufficient, whereas subsidies are required when carbon-market revenues are constrained by physical issuance capacity or aggregate credit supply. The analysis further shows that intervention boundaries and carbon price dynamics materially affect policy efficiency, project solvency, and bond valuation. The paper provides a structural explanation of the greenium by linking externality internalization, policy-induced cash-flow repair, and bond-level option value, offering guidance for adaptive green finance policy design.

Keywords: Green investment, Externality internalization, Carbon trading, Government subsidy, Decision optimization, Green bond pricing

1. Introduction

Sustainable development and environmental protection are global priorities, requiring effective financial instruments to fund green projects and facilitate the transition to a green economy (Starks, 2023). Green finance has emerged as an important channel through which private capital can support

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environmental protection, low-carbon infrastructure, and renewable energy investment. Within this system, green bonds play a particularly important role because they link capital-market financing to projects with explicit environmental objectives, thereby helping issuers channel investment toward activities that generate positive environmental externalities (Flammer, 2020, 2021). However, many green projects remain financially unattractive because their private cash flows do not fully reflect the social value of emission reductions and other environmental benefits. This wedge between private and social returns creates an underinvestment problem in low-carbon technologies and climate-related infrastructure (Abrell et al., 2019; Rennings, 2000; Jaffe et al., 2002). The pricing of green bonds therefore depends not only on the green label itself, but also on whether environmental benefits can be converted into cash flows that are measurable, credible, and relevant to investors. When emission-reduction benefits remain unpriced or only imperfectly monetized, green projects may exhibit weak standalone profitability, raising financing costs and limiting the depth of green bond markets (Zerbib, 2019; Larcker and Watts, 2020; Fatica et al., 2021; Li et al., 2022).

This externality-pricing friction is reflected in the still-limited penetration and depth of green bond markets. Larcker and Watts (2020) identify higher issuance costs as a primary barrier, attributing this to the emission reduction costs and unpriced positive externalities inherent in green projects (Han et al., 2024). Without effective compensation mechanisms, these increased costs distort green bond pricing and impede optimal resource allocation (Abrell et al., 2019). This issue is particularly relevant for China. To achieve its dual carbon goals, peaking emissions by 2030 and reaching carbon neutrality by 2060 (Yin et al., 2024), the government has vigorously cultivated its domestic green bond market. In 2024, the domestic market issued 477 green bonds totaling 681.43 billion yuan.[1] However, these figures represented only 0.85% of the total bond volume and 0.93% of total issuances. Compared with 2023, the primary market contracted, with secondary market liquidity and participation still lagging. These facts further highlight the need to internalize environmental externalities into cash flows that can support project viability and be recognized in bond pricing.

Green bonds are specialized instruments for funding projects with positive externalities that current pricing mechanisms often fail to capture, resulting in divergences between social and private returns and an underinvestment in green innovation (Naeem et al., 2021; Flammer, 2020, 2021). While investors increasingly integrate environmental, society, and governance (ESG) principles into their portfolios, seeking to balance sustainability outcomes with financial returns (Huynh and Xia, 2021; Berk and Van Binsbergen, 2025; Bond and Levit, 2025), the public-goods nature of ecological systems still necessitates policy intervention to achieve Pareto efficiency in environmental governance (Ding et al., 2023; Wiskich, 2024). Such efforts, whether market- or policy-driven, must be evaluated through a comprehensive cost-benefit analysis (Zhang et al., 2024; Pedersen et al., 2021).

The internalization of carbon externalities constitutes a fundamental rationale for environmental regulation, China's carbon emissions trading scheme (ETS) is currently transitioning toward market-oriented mechanisms (Wang and Zhang, 2022; Jia and Lin, 2020). Unlike the EU's fixed-cap system, China's ETS incorporates output-based features in which allowances are linked to actual production activity (Jotzo et al., 2018; Zhang and Shi, 2025), aiming to achieve emission targets at lower social

[1] See `http://ccxgf.com.cn/`

cost (Anke et al., 2020; Pang and Timilsina, 2021). Within this context, the strategic design of quota allocation serves as a primary lever for driving innovation (Zhang et al., 2022). Therefore, optimization models are widely employed to identify optimal strategies under constraints (Pu et al., 2023; Ren et al., 2022; Wang et al., 2025a). For instance, Zhu et al. (2023) and Xia et al. (2023) developed bi-level and multi-objective models to balance cost-efficiency and fairness in quota distribution, while others have applied optimization to minimize footprints in transport and production scheduling (Yin et al., 2021; Trevino-Martinez et al., 2022).

However, since China's ETS remains in its early stages and faces challenges such as regional disparities, uncertain price signals, and inconsistent quota allocation (Li et al., 2025; Gao et al., 2020; Song et al., 2022), carbon trading alone may not be sufficient to incentivize large-scale green investment. Fiscal subsidies therefore emerge as a critical policy backstop to correct market distortions and relax financing constraints (Gugler et al., 2021; Zhang and Zhang, 2021; Zhong et al., 2023). To evaluate these interventions, Liu et al. (2025a) utilized the general equilibrium model to assess the impacts of heterogeneous green subsidy policies. Yang et al. (2020) validated the superiority of the externality internalization approach by explicitly incorporating externality costs into subsidy optimization. Beyond lowering investment barriers (Cao et al., 2020; Liu et al., 2025b; Bian and Zhao, 2020), dynamically adjusted subsidies can effectively optimize resource allocation (Ye et al., 2021; Bialek and Ünel, 2022; Fan and Chen, 2024). Hussain and Lee (2022) emphasize that policy matrices combining carbon quotas and subsidies are more effective for externality internalization than single instruments (Yin et al., 2024). To maximize these synergies, Martelli et al. (2020) and Hu et al. (2022) developed models to coordinate fiscal incentives with multi-factor constraints (Wang et al., 2025b). This evolution toward synergistic policy mixes provides the theoretical foundation for the dynamic optimization framework proposed in this study.

Green bonds share fundamental financial attributes, such as credit and liquidity risks, with conventional bonds (Dong et al., 2021; Agliardi and Agliardi, 2021, 2019), yet they are distinguished by environmental externalities that significantly influence investor demand and pricing (Zeng et al., 2023). Empirical evidence on the greenium remains mixed (Karpf and Mandel, 2018; Baker et al., 2022; Fatica et al., 2021; Zerbib, 2019; Larcker and Watts, 2020). At the same time, responsible investing frameworks increasingly incorporate environmental preferences, carbon risk, and climate-transition exposure into asset pricing (Avramov et al., 2022; Bolton and Kacperczyk, 2021, 2023; Döttling and Rola-Janicka, 2025). Specifically, Feldhütter et al. (2024) decompose this premium into conventional cash flows and ESG-linked optionalities, reflecting investors' acceptance of lower returns in exchange for non-monetary benefits (Pástor et al., 2021; Caramichael and Rapp, 2024; Feldhütter and Pedersen, 2024; Bonnefon et al., 2025). In the Chinese market, Li et al. (2022) suggests that policy interventions, rather than investor preferences alone, remain a primary determinant of the green premium.

Despite extensive analysis of policy optimization and green bond premiums, financial research often treats environmental interventions as exogenous shocks, overlooking the feedback mechanisms on asset pricing. Furthermore, analyzing policy instruments in isolation obscures the synergistic potential and substitution effects between market-based and fiscal tools. Therefore, three questions remain open. First, how can externality internalization be incorporated into a tractable policy-

optimization framework? Second, how should heterogeneous policy instruments be coordinated to restore project viability while preserving fiscal discipline? Third, how do these instruments affect green projects' investment attractiveness and green bond pricing?

To address these questions, this paper proposes a two-stage model that endogenizes the transmission mechanism from macro-level policy optimization to micro-level asset pricing. Moving beyond ex-post evaluation (Zhang and Fan, 2024), the first stage provides an ex-ante decision framework for allocating carbon credits and fiscal subsidies so as to restore project viability while minimizing the lifecycle commitment of policy resources. The model embeds a market-first intervention hierarchy, where carbon-credit revenues are prioritized as the primary channel for monetizing emission reductions, while subsidies are used as a fiscal backstop when carbon-market revenues are insufficient. The second stage maps the optimized policy-adjusted cash flows into an option-pricing framework. This integration shows how policy support repairs downside cash-flow shortfalls, shifts green assets from a distressed state toward financial viability, and allows uncertainty in future environmental value to be priced as upside potential. The framework therefore provides a structural rationale for the greenium, by linking externality internalization, project cash-flow repair, and bond-level valuation.

This paper contributes to the literature in four ways. First, it bridges environmental policy optimization and green bond pricing, by modelling the endogenous transmission from policy instruments to project cash flows and, ultimately, to bond-level valuation. This differs from studies that treat policy support as an exogenous shock or estimate greenium directly from observed yield differentials. Second, it develops a synergistic policy-allocation framework that coordinates carbon credits and fiscal subsidies under credit-supply and budget constraints. By distinguishing the revenue-side role of carbon credits from the cost-side role of subsidies, the framework clarifies how market-based and fiscal instruments jointly restore project viability while preserving fiscal discipline. Third, it provides a structural explanation of the greenium through an option-pricing channel. Optimized policy intervention repairs downside cash-flow shortfalls and shifts green assets from a distressed state toward financial viability, allowing uncertainty in future environmental value to be capitalized as upside optionality. Finally, it analyzes how intervention boundaries and carbon price dynamics affect policy effectiveness, project viability, and bond pricing, offering guidance for the design of adaptive green policies.

The rest of this paper is organized as follows. Section 2 details model construction. Sections 3 and 4 present the empirical results for policy optimization and green bond pricing, respectively. Section 5 and Section 6 provide an extension to the baseline model, including robustness checks and dynamic analysis. Section 7 discusses the economic mechanisms and the market implications. Finally, Section 8 concludes.

2. The Model

This section introduces the proposed two-stage framework that links policy intervention to green bond pricing. The first stage models project-level revenues and costs and solves for the optimal allocation of carbon credits and fiscal subsidies required to restore financial viability. The second stage maps the optimized policy-adjusted project cash flows into a bond-level pricing model, where

the embedded green attribute is valued as an option-like claim created by the internalization of environmental externalities.

2.1. Revenue-cost modelling of green projects

Let us consider a portfolio of green bonds, indexed by $i \in \{1, 2, ..., I\}$. Each bond i finances a set of green projects indexed by $j \in J_i$, where J_i denotes the project pool associated with bond i. The green project (i, j) operates over periods $l \in \{1, 2, ..., L_{i,j}\}$. The model tracks the revenue, cost, carbon-credit allocation, and subsidy intervention for each project-year pair. The conventional operating revenue from product (i, j) in period l is:

$$R_{i,j,l}^{prod} = p_{i,j,l} q_{i,j,l}, \tag{1}$$

where $p_{i,j,l}$ and $q_{i,j,l}$ are the product price and quantity, respectively.

The carbon-market component is determined by the difference between the carbon credits allocated to the project and its net emissions (Yin et al., 2024; Shan et al., 2024). More specifically, let $v_{i,j,l}^{b}$ denote the carbon credits allocated to project (i, j) in period l, $v_{i,j,l}$ denote the baseline emission level, and $a_{i,j,l}$ denote the emission reductions achieved by the project. The net carbon-market revenue is

$$R_{i,j,l}^{carbon} = p_l^c \left[v_{i,j,l}^{b} - (v_{i,j,l} - a_{i,j,l}) \right], \tag{2}$$

where p_l^c is the carbon price in period l. In our baseline analysis, p_l^c is assumed to be a constant over time, whereas in the dynamic analysis, it is allowed to evolve over time to capture alternative carbon-market trajectories.

The total revenue of project (i, j) in period l is therefore defined as the combination of conventional product revenue and net income from carbon trading:

$$R_{i,j,l} = p_{i,j,l} q_{i,j,l} + p_l^c \left[v_{i,j,l}^{b} - \big(v_{i,j,l} - a_{i,j,l}\big) \right]. \tag{3}$$

On the cost side, the project incurs upfront investment, ongoing operating expenses, and emission reduction costs. Let $f_{i,j,l}$ denote the fixed investment cost, incurred only at the initial period ($l = 1$), $c_{i,j,l}$ denote the unit operating cost, and $\omega_{i,j,l}$ denote the emission-reduction cost coefficient. Let $\chi_{i,j,l} = a_{i,j,l}/v_{i,j,l}$ denote the emission reduction rate. The reduction cost is assumed to be a quadratic function of the reduction level, $\frac{1}{2}\omega_{i,j,l} q_{i,j,l} \chi_{i,j,l}^2$, reflecting increasing marginal reduction costs (Wang et al., 2022; Zhang and Shi, 2025). Hence, the base cost before subsidy is

$$K_{i,j,l}^{base} = f_{i,j,l} \cdot \mathbb{I}_{\{l=1\}} + c_{i,j,l} q_{i,j,l} + \frac{1}{2} \omega_{i,j,l} q_{i,j,l} \chi_{i,j,l}^2, \tag{4}$$

where $\mathbb{I}_{\{l=1\}}$ is an indicator function equal to 1 in the initial period and 0 otherwise.

Concurrently, the government may subsidize a fraction of project cost. Let $k_{i,j,l} \in [0, 1]$ denote the subsidy coefficient (Hussain et al., 2022). The subsidy amount for project (i, j) at time l is given by:

$$Subsidy_{i,j,l} = k_{i,j,l} K_{i,j,l}^{base}, \tag{5}$$

and the total cost after the government subsidy is

$$K_{i,j,l} = (1 - k_{i,j,l}) \left(f_{i,j,l} \cdot \mathbb{I}_{\{l=1\}} + c_{i,j,l} q_{i,j,l} + \frac{1}{2} \omega_{i,j,l} q_{i,j,l} \chi_{i,j,l}^2 \right). \tag{6}$$

Eqs. (3) and (6) define the cash-flow structure for project (i, j) at time l that enters the optimization problem. Carbon credits affect project viability through the revenue side, while subsidies affect viability through the cost side. This distinction is central to the later transmission from policy intervention to green bond pricing.

2.2. The optimization model

This section proposes an optimization model to determine the optimal government resource allocation strategy. The objective function contains two components, weighted by a parameter $\mu \in [0, 1]$. The first component, representing the discounted policy-preference cost, measures the present value of policy interventions adjusted by regulatory priorities. The second component, termed nominal policy resource commitment, addresses the non-discounted scale of policy resources utilized over the project lifecycle. More specifically,

$$\begin{aligned} \min_{\{v_{i,j,l}^b, k_{i,j,l}\}} \quad & \underbrace{\mu \sum_{i=1}^{I} \sum_{j \in J_i} \sum_{l=1}^{L_{i,j}} \frac{\lambda_1 p_l^c v_{i,j,l}^b + \lambda_2 Subsidy_{i,j,l}}{(1+r)^l}}_{\text{Discounted policy-preference cost}} + \\ & \underbrace{(1-\mu) \sum_{i=1}^{I} \sum_{j \in J_i} \sum_{l=1}^{L_{i,j}} \left(p_l^c v_{i,j,l}^b + Subsidy_{i,j,l} \right)}_{\text{Nominal policy resource commitment}}. \end{aligned} \tag{7}$$

where r is the discount rate and coefficients (λ_1, λ_2) represent the policy preference weights for carbon credits and subsidies. The parameter μ acts as an objective trade-off weight, governing the trade-off between minimizing discounted policy costs and minimizing nominal policy resource commitments.

To operationalize the market-first intervention hierarchy, where carbon credits serve as the primary support mechanism and subsidies remain supplementary (Zhang and Fan, 2024), we set $\lambda_2 \gg \lambda_1$. This gives priority to carbon-credit support when market-based environmental revenues can restore project viability, while allowing subsidies to operate as a residual fiscal backstop when carbon credits are insufficient. This mechanism effectively prevents the substitution effect where subsidies might otherwise crowd out market-driven solutions.

This dual-target objective disciplines both the present-value cost and the lifecycle scale of policy intervention. The discounted policy-preference cost term identifies intervention paths that restore project viability at the lowest present-value cost to the regulator. The nominal policy resource commitment term penalizes the undiscounted resource allocation associated with carbon credits and fiscal subsidies over the project life. Economically, this second component prevents the model from treating carbon credits as an unlimited policy instrument and reduces the risk of excessive credit injection that could weaken carbon-market price signals. It also limits the cumulative fiscal burden imposed on the public budget. Mathematically, the non-discounted component helps distinguish and

select an economically meaningful allocation among alternative intervention paths that may deliver similar discounted costs. The resulting solution is therefore less sensitive to arbitrary timing choices and better reflects the scarcity of both market-based and fiscal policy resources.

The model employs two decision variables for each project (i, j) at each time period l:

- $v^b_{i,j,l}$: The volume of carbon credits allocated to project (i, j) at time l, representing a quantitative, market-based instrument.
- $k_{i,j,l}$: The subsidy coefficient for project (i, j) at time l, representing a price-based, fiscal instrument.

To provide sufficient but not excessive incentives, the net present value (NPV) of each project's cash flow is constrained to be zero, which ensures policy support offset the financing gap created by unpriced environmental externalities, while preventing overcompensation. Although regulators may face informational and implementation frictions in targeting exact break-even outcomes in practice, this equality serves as a theoretical benchmark for cost-efficient intervention. It characterizes a social planner who aims to internalize externalities without creating excess project rents. In Subsection 5.1, we further examine a relaxed viability constraint, $NPV_{i,j} \geq 0$, to test whether the main results are driven by the equality restriction.

For each project (i, j), the constraint is formulated as:

$$NPV_{i,j} = \sum_{l=1}^{L_{i,j}} \frac{R_{i,j,l} - K_{i,j,l}}{(1+r)^l} = 0, \quad \forall i, j. \tag{8}$$

Furthermore, the total volume of carbon credits allocated across all projects at any time l cannot exceed the government's available carbon credit cap G_l (Shan et al., 2024):

$$\sum_{i=1}^{I} \sum_{j \in J_i} v^b_{i,j,l} \leq G_l, \quad \forall l. \tag{9}$$

Similarly, the total subsidy across all projects at time l is constrained by

$$\sum_{i=1}^{I} \sum_{j \in J_i} Subsidy_{i,j,l} = \sum_{i=1}^{I} \sum_{j \in J_i} k_{i,j,l} K^{base}_{i,j,l} \leq B_l, \quad \forall l, \tag{10}$$

where B_l denotes the government's direct subsidy budget in period l.

Finally, decision variables are restricted to non-negative values, defining the government's role as a provider of support. To maintain reduction incentives and prevent over-support, the credit allocation shall not exceed baseline emissions, and the subsidy coefficient is capped. More specifically,

$$0 \leq {v^b}_{i,j,l} \leq v_{i,j,l}, \quad \forall i, j, l, \tag{11}$$

and

$$0 \leq k_{i,j,l} \leq 1, \quad \forall i, j, l. \tag{12}$$

Collectively, the constraints in Eqs. (8)–(12) establish the structural boundaries of the optimization framework, integrating fundamental economic principles with regulatory capacity. At its core, the break-even constraint in Eq. (8) operationalizes a just enough principle, ensuring that government support precisely offsets the financial deficit caused by externalities without creating excess profits. The resource limits in Eq. (9)–(10) impose macro-level fiscal and regulatory discipline. Finally, Eqs. (11)–(12) formally define the government's role as a provider of one-way support, guaranteeing that all instruments function as positive economic and reduction incentives.

By executing this optimization model and internalizing externalities into the project's solvency boundary, our framework transforms macro-level policy instruments into micro-project cash flows, which was often treated as exogenous shock in previous studies. The resulting optimal allocation set, $\left\{v_{i,j,l}^{b*}, k_{i,j,l}^{*}\right\}$, represents a dynamic and differentiated strategy. By determining the optimal timing and magnitude of interventions, this strategy provides a quantitative guide to maximize incentives for green projects while maintaining strict fiscal prudence.

2.3. The pricing model

This section integrates the optimal policy intervention, $\left\{v_{i,j,l}^{b*}, k_{i,j,l}^{*}\right\}$, derived from the optimization stage in Subsection 2.2 into green bond pricing. For clarity, all variables and parameters are summarized in Table 1 before formal pricing. Building on the decomposition framework proposed by Hu and Tian (2024), we conceptualize the green bond's total value as the sum of a conventional fixed-income component and a green attribute component. While the former is valued through the standard discounted cash flow (DCF) analysis, the latter represents an embedded call option on the project's potential for externality-to-asset conversion.

To isolate the value of this green attribute, we assume a stylized special purpose vehicle (SPV) structure to finance the green project (Assab, 2025). The conceptual legal and cash flow structure of this SPV-based financing is illustrated in Fig. 1. This setup renders the green bond independent of the parent company's balance sheet, ensuring that the valuation reflects only the project's intrinsic cash flows and the localized impact of carbon credits and subsidies.[2] Besides, the SPV, as the project owner, is legally entitled to register verified emission reductions as CCERs and trade them on the national exchange, thereby providing a direct link between environmental performance and financial cash flow.[3] Finally, the SPV structure and the policy interventions allow our analysis to abstract from issuer default risk (Barrieu and Fehr, 2014; Zhang and Fan, 2024), enabling us to focus exclusively on the greenium generated by underlying volatility and policy support rather than by the issuer's solvency.

Because a bond typically finances only a share of total project investment, we attribute project-

[2]While the SPV's cash flows are prioritized for debt service to protect bondholders, the parent corporation, as the residual equity holder, benefits from the project's net profits and the environmental value created after the bond's maturity. This structure aligns the incentives of the corporation with the environmental performance of the project.

[3]See the Ministry of Ecology and Environment of the People's Republic of China, `https://www.mee.gov.cn/xxgk2018/xxgk/xxgk02/202310/t20231020_1043694.html`.

Table 1: Parameters and variables with standardized units

Parameter/Symbol	Definition	Unit
Indices		
i	Bond index, $i \in \{1, \ldots, I\}$	-
j	Project index, $j \in \{1, \ldots, J_i\}$	-
l	Time period index for projects, $l \in \{1, \ldots, L_{i,j}\}$	-
t	Time period index for bonds, $t \in \{1, \ldots, T_i\}$	-
Optimization and Policy Parameters		
μ	Objective trade-off weight	-
λ_1, λ_2	Policy preference weights for credits and subsidies	-
p_l^c	Carbon trading price	CNY/tons
G_l	Government's available carbon credit cap at time l	million tons
B_l	Government's direct subsidy budget at time l	million CNY
Financial and Project Parameters		
r	Discount rate	-
σ	Volatility of the asset value	-
$\phi_{i,j}$	Financing contribution ratio ($Y_{i,j}/X_j$)	-
$Y_{i,j}, X_j$	Bond financing and Total project investment	million CNY
$L_{i,j}, T_i$	Project operational lifespan and Bond maturity	Year
$p_{i,j,l}, q_{i,j,l}$	Product sales price and Demand quantity	CNY/MWh, MWh
$a_{i,j,l}, v_{i,j,l}$	Emission reduction and Baseline carbon emissions	million tons
$f_{i,j,l}, c_{i,j,l}$	Fixed cost and Unit production cost	million CNY, CNY/MWh
$\omega_{i,j,l}$	Emission reduction cost coefficient	CNY/MWh
$\chi_{i,j,l}$	Emission reduction rate ($a_{i,j,l}/v_{i,j,l}$)	-
Option Pricing Variables		
S_i	Pro-rated revenue (Underlying asset value)	million CNY
K_i	Pro-rated cost (Option strike price)	million CNY
C_i	Value of the embedded green option	million CNY
Decision Variables		
$v_{i,j,l}^b$	Carbon credits allocated to project	million tons
$k_{i,j,l}$	Government subsidy coefficient	-

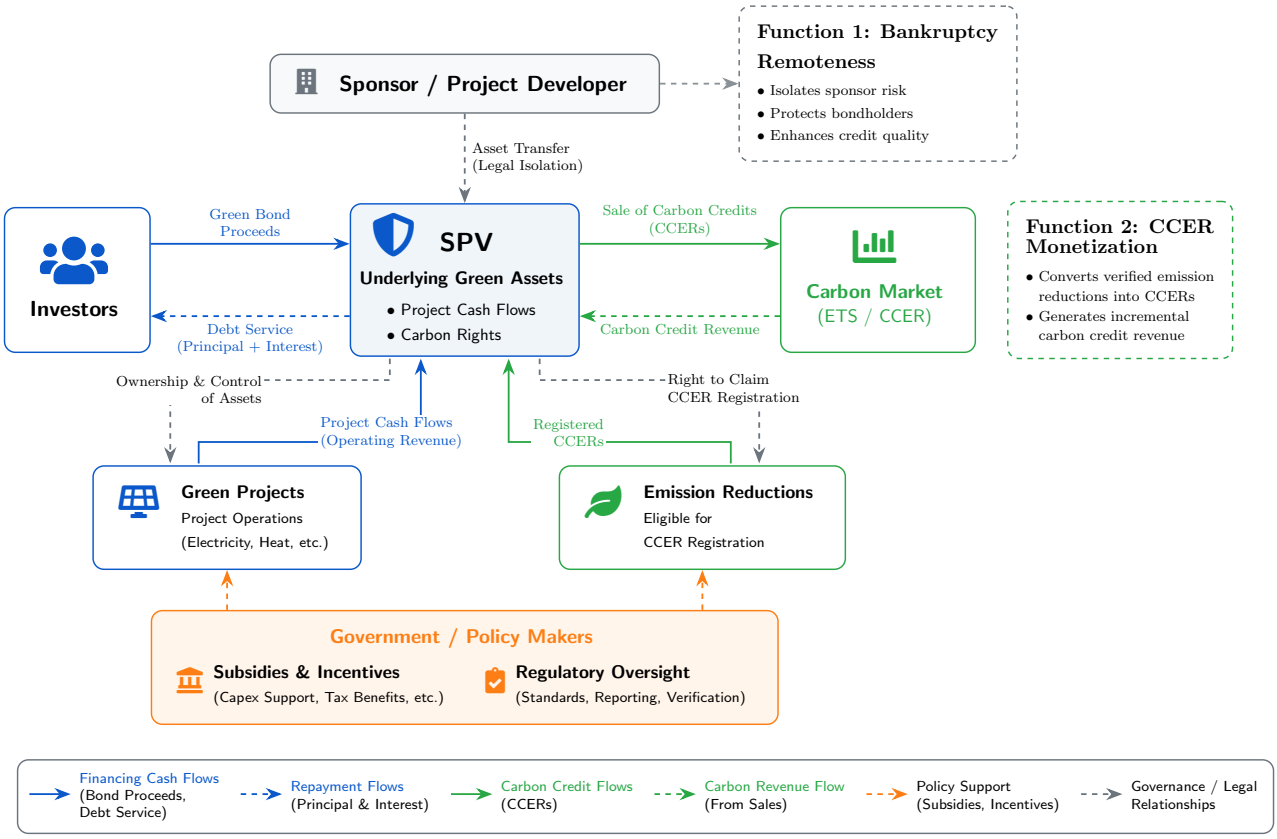


Figure 1: The SPV structure of a green bond

level cash flows to bond based on the pro-rata financing share:

$$\phi_{i,j} = \frac{Y_{i,j}}{X_j}, \tag{13}$$

where $Y_{i,j}$ represents the amount of financing provided by bond i to project j, and X_j denotes the total investment in project j. Project-level cash flows and environmental benefits are evaluated over the operational lifespan of each project, $L_{i,j}$, and are then attributed to bond i according to its pro-rata financing share, $\phi_{i,j}$. This ensures that the green premium embedded in the bond pricing reflects only the specific share of environmental value attributed to the bondholders.

The pro-rated revenue-side asset value for bond i, S_i, is

$$S_i = \sum_{j \in J_i} \phi_{i,j} \sum_{l=1}^{L_{i,j}} \frac{1}{(1+r)^l} \left\{ p_{i,j,l} q_{i,j,l} + p_l^c \left[v_{i,j,l}^{b*} - (v_{i,j,l} - a_{i,j,l}) \right] \right\}. \tag{14}$$

The corresponding pro-rated cost-side benchmark, K_i, for bond i is

$$K_i = \sum_{j \in J_i} \phi_{i,j} \sum_{l=1}^{L_{i,j}} \frac{1}{(1+r)^l} (1 - k_{i,j,l}^*) \left(f_{i,j,l} \mathbb{I}_{\{l=1\}} + c_{i,j,l} q_{i,j,l} + \frac{1}{2} \omega_{i,j,l} q_{i,j,l} \chi_{i,j,l}^2 \right). \tag{15}$$

To quantify the value of the green attribute, we employ an option-pricing framework that captures the intertemporal asymmetry in green projects and upward optionality created by policy interventions (Chen and Yang, 2023; Bertrand, 2024; Ye et al., 2025). The economic intuition is that project costs are relatively more predictable once investment and operating plans are fixed, whereas the benefits from environmental performance are long-dated and exposed to upside potential from carbon price appreciation, technology scaling, and future emission-reduction value. In this setting, the revenue-side asset value, S_i, represents the present value of bondholder-attributed project revenues and internalized environmental benefits. The cost-side benchmark, K_i, represents the present value of the cost threshold that must be covered for these green benefits to become financially realizable.

The green option captures the long-term upside of environmental growth while the downside is capped. Policy intervention changes the state of this embedded option by repairing the project cash-flow shortfall identified in the first-stage optimization. When the project remains financially distressed, the embedded green option is out-of-the-money (OTM) because $S_i < K_i$. Once carbon-credit revenues and subsidy support restore viability, the option moves toward an at-the-money (ATM) state, where upside uncertainty in future environmental value becomes priced rather than merely risky. This interpretation acknowledges that the financial appeal of green bonds stems from their ability to capture long-term environmental premiums, while the option value serves as a contingent claim on the project's successful internalization of externalities. Therefore, the option value C_i captures the environmental value accruing to the bondholders, effectively isolating the green attribute's contribution to the total bond price, as follows:

$$C_i = S_i N(d_1) - K_i N(d_2),$$

where

$$d_1 = \frac{\ln(\frac{S_i}{K_i}) + \frac{\sigma^2}{2}T_i}{\sigma\sqrt{T_i}}, d_2 = \frac{\ln(\frac{S_i}{K_i}) - \frac{\sigma^2}{2}T_i}{\sigma\sqrt{T_i}}.$$

Here $N(\cdot)$ denotes the standard normal cumulative distribution function, σ denotes the volatility of the project asset value associated with bond i, and T_i the option maturity.[4]

3. Empirical setting and optimal policy analysis

3.1. The setting and data description

To empirically validate the proposed framework, we apply the model to the Chinese green bond market. We conduct a case study on the Chinese clean energy sector, focusing on Jilin Electric Power Co., Ltd. (Jidian).[5] Since 2022, Jidian has issued three tranches of carbon-neutral green corporate bonds totaling 2.5 billion CNY. These bonds, certified against both national and carbon-neutral standards, have consistently maintained the highest green rating (G1).[6] As a state-owned enterprise, Jidian exhibits strong policy alignment and relatively high credit stability, which is typically associated with low default risk in the Chinese corporate bond market (Zhang and Fan, 2024). This institutional feature supports the assumption of negligible default risk in our framework, allowing us to focus on the valuation of green attributes. Essential issuance characteristics of these bonds are summarized as follows:

22 Jidian G1: Issued in March 2022, with a principal amount of 1 billion CNY, a 5-year maturity, and a 3.80% coupon rate. The proceeds are primarily allocated to fund wind power projects.

22 Jidian G2: Issued in October 2022, with a principal amount of 1 billion CNY, a 3-year maturity, and a 2.90% coupon rate. The proceeds were used to fund a portfolio of clean energy projects.

23 Jidian GCKV01: Issued in December 2023, with a principal amount of 0.5 billion CNY, a 3-year term, and a 2.99% coupon rate. The proceeds were mainly used to refinance existing clean energy project loans.

The bond proceeds financed a portfolio of wind and photovoltaic (PV) projects, as shown in Appendix B. All project-level data were retrieved from official bond prospectuses and third-party certification reports. With these data, we calibrate two groups of model parameters: project-level operating parameters and carbon-related financial parameters. Unless otherwise stated, all monetary values are denominated in CNY.

[4] To quantify the green attribute's value, we model the optimized cash flows as a call option. Following Mason and Merton (1985) and Copeland and Antikarov (2003), we assume that the capital value of the green attribute (S_i) follows a Geometric Brownian Motion (GBM) (Hu and Tian, 2024). Since S_i and K_i are the discounted present values of revenues and costs respectively, we utilize the present value form of the Black-Scholes model (Barrieu and Fehr, 2014), where the parameters d_1 and d_2 are simplified by neutralizing the explicit interest rate term to ensure mathematical consistency and avoid double-discounting.

[5] In August 2025, Jilin Electric Power Co., Ltd. officially updated its corporate name to SPIC Green Energy Development Co., Ltd. (SPIC GED). Per company announcement (No. 2025-065), the legal entity remains unchanged, and all existing contracts and obligations under the former name continue to be binding. This administrative change does not affect the data consistency or the research findings.

[6] Here, G1 denotes the green rating, signifying the highest level. It is important to distinguish this from the G1 appearing in the bond name 22 Jidian G1.

Project-Level Parameters. The average on-grid tariff ($p_{i,j,l}$) is set at 467.70 CNY/MWh for wind and 555.29 CNY/MWh for PV, with corresponding unit operating costs ($c_{i,j,l}$) of 217.81 and 266.91 CNY/MWh, respectively. Derived from the cost differential between renewable and benchmark technologies, the emission reduction cost coefficient ($\omega_{i,j,l}$) is 12.89 CNY/MWh for wind and 61.99 CNY/MWh for PV. To maintain modeling parsimony, these values are assumed to remain constant throughout the project lifecycles.

Besides, wind and PV projects exhibit zero direct operational emissions, distinguishing them from traditional emission-intensive industrial sectors. To align with this institutional reality, we redefine baseline emissions ($v_{i,j,l}$) as counterfactual emissions, representing the CO_2 emissions that would have occurred if the same amount of electricity were supplied by the thermal-dominated grid. Consequently, the project's emission reduction is defined as the avoided grid emissions relative to this baseline ($v_{i,j,l} = a_{i,j,l}$). Under this framework, the allocated carbon credits ($v^b_{i,j,l}$) are interpreted within the framework of China Certified Emission Reductions (CCERs).

To evaluate the inherent profitability and investor attractiveness, the discount rates are set at 14.81% for wind and 11.83% for PV. These rates are proxied by the internal rates of return (IRR) of representative projects, reflecting the required returns for investors.

Carbon-Related Financial Parameters. To reflect the market environment at bond issuance, we set 2022 as the base year for carbon pricing. Following Hussain et al. (2022), the carbon price, $p^c_l = 70$ CNY/ton and remains constant in the baseline specification to isolate allocation dynamics from carbon price fluctuations, ensuring that results are driven by the allocation mechanism rather than price uncertainty. In the dynamic extension, p^c_l is allowed to vary over time to capture alternative carbon price trajectories and assess robustness under time-varying price paths.

The asset volatility σ represents the annual fluctuation of the project's underlying value. Since project-level value is not directly observable, we calibrate a proxy based on market data. While the parent company's equity volatility (σ_E) provides a benchmark, it systematically overestimates specific project risk due to financial leverage and business diversification. To isolate project-specific risk, we derive the unlevered asset volatility (σ_A) using the standard deleveraging formula: $\sigma_A = \sigma_E \times [E/(D+E)]$, where $E/(D+E)$ represents the equity-to-asset ratio. Based on the issuer's 69.57% leverage and historical equity volatility of σ_E=35.12%, we derive an implied asset volatility of σ_A=10.69%.[7]

The empirical analysis in this section is designed to address the following research questions:

RQ1: How can government policy instruments be optimally allocated under constraints to internalize positive externalities and ensure the financial viability of green projects?

RQ2: How do these optimized policy interventions transmit to the financial market to determine green bond pricing, and how is the value of volatility integrated into the asset valuation?

RQ3: How should the policy design dynamically adapt to variations in the external economic

[7]To validate the reasonableness of our volatility parameter, we calculated the unlevered asset volatility of China Longyuan Power Group, the world's largest wind power operator and a leading pioneer in China's renewable energy sector. Its calculated unlevered asset volatility is σ =11.24%, which aligns closely with our calibrated baseline of σ =10.69%. To ensure the robustness, we also conducted sensitivity tests across the ±20% range, see Appendix A.3.

environment to maintain policy effectiveness?

3.2. Optimization results

We conduct a numerical analysis to empirically validate the proposed framework and investigate the distinct roles of market-based mechanisms and policy interventions. The analysis highlights the necessity of a coordinated policy mix in achieving efficient externality internization and financially viable green investment outcomes. Specifically, we analyze the following scenarios:

(1) Baseline scenario: no carbon market or zero carbon credits (i.e., p_l^c=0 or v^b=0);

(2) Unconstrained optimal credit allocation: optimization with unlimited carbon credits;

(3) Constrained optimal credit allocation: optimization under annual credit caps without subsidy;

(4) Synergistic optimization: optimal combination of credits and subsidies under dual constraints.

3.2.1. Scenario 1: Baseline scenario

First, we simulate the baseline scenario by setting the carbon price $p_l^c = 0$ or the allocated credits v^b=0.[8] In this scenario, project revenue originates exclusively from electricity sales. The baseline results are reported in Column 2 of Table 2, which consistently shows a negative NPV in all projects. Table 2 further translates these project-level deficits into the carbon-credit quantities required under alternative allocation regimes. Columns 3–8 report the physical quantities of carbon credits required to generate a discounted present value equal to the corresponding NPV deficit in Column 2. This outcome validates our hypothesis that conventional electricity revenues are insufficient to cover the high upfront capital expenditures and operational costs of green projects. This significant deficit, primarily attributable to unpriced positive environmental externalities, directly constrains the deployment of clean energy technologies. Consequently, such projects remain inherently unprofitable and fail to attract private investment, highlighting a market failure and underscoring the imperative for policy intervention to stimulate sustainable development.

Table 2: Required carbon credit v^b under different constraints

Project	NPV Deficit (million CNY)	Required v^b values in million tons (Mt)					
		Unconstrained (Discounted)	Unconstrained (Dual-obj.)	$G_l = 8$	$G_l = 7$	$G_l = 6$	$G_l = 5.5$
G1-Wind-A	727.86	164.64	11.94	13.19	13.51	15.94	16.62
G1-Wind-B	1172.40	265.19	19.23	24.55	26.61	28.18	30.74
G2-Wind	248.07	56.11	4.07	4.07	4.07	4.07	4.07
G2-PV-A	30.84	7.21	0.49	0.96	1.21	2.11	4.61
G2-PV-B	850.17	198.78	13.58	24.33	31.99	43.65	67.49
G3-PV	307.70	71.94	4.92	8.71	8.37	15.12	21.81
Total	**3337.04**	**763.87**	**54.23**	**75.81**	**85.76**	**109.07**	**145.34**

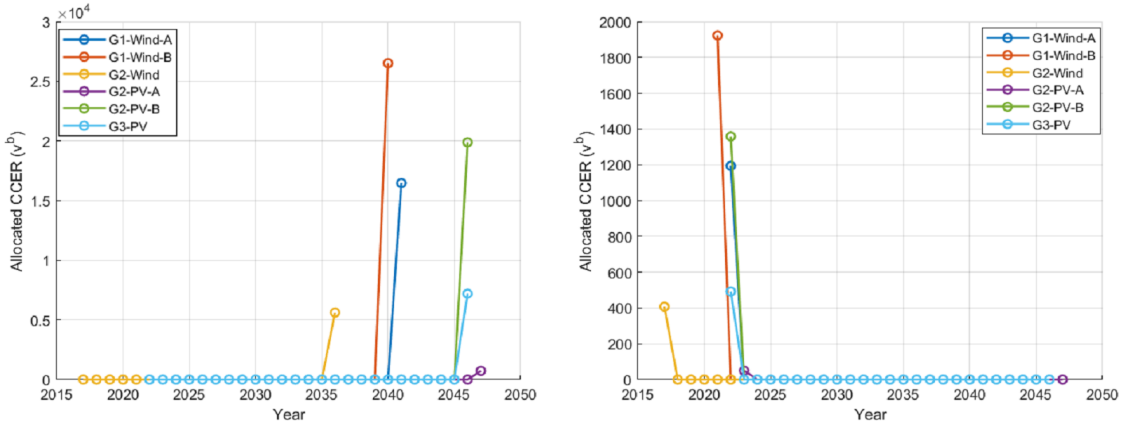
Note: Column 2 reports the baseline NPV deficit for each project. Columns 3–8 report the physical carbon-credit quantities, measured in million tons, required under alternative allocation regimes to generate a discounted present value equal to the corresponding deficit in Column 2. Although the physical volumes of v^b vary across allocation regimes, each allocation yields the same present-value compensation and offsets the aggregate financial gap, i.e., NPV(v^b) = Deficit = 3337.04 million CNY.

[8]Since clean energy projects have zero operational emissions, these two settings are mathematically equivalent in our model. But for traditional emission enterprises, these two settings need to be discussed separately.

3.2.2. Scenario 2: Unconstrained optimal credit allocation

The second scenario investigates the impact of introducing the carbon market ($p_l^c > 0$). In this setting, we activate the optimization framework while assuming unlimited carbon credit ($G_l = \infty$). Given the market-first hierarchy and unlimited carbon credit availability, subsidies are not activated. The objective function is reformulated to minimize the value of carbon credits.

To evaluate the role of the nominal policy resource commitment component in the dual-target objective in Eq. (7), we compare two unconstrained allocation schemes: discounted-cost minimization only and dual-objective minimization. The first scheme, reported in Column 3 of Table 2 and illustrated in Fig. 2(a), excludes the penalty on nominal policy resource commitment. Under this specification, the solution finds a feasible allocation that exactly offsets the aggregate NPV deficit of 3337.04 million CNY. However, because later credits receive lower present-value weights, majority of credits are allocated toward the later operational years, particularly 2036–2047. This back-loaded timing substantially increases the required physical quantity of credits, with the total nominal volume rising to 763.87 million tons, far above the volume required under the dual-objective specification in Column 4 of Table 2. The result shows that discounted-value minimization alone can satisfy the present-value constraint but does so through an economically inefficient allocation path. Since multiple time profiles can generate the same discounted value, an optimization criterion based only on discounted cost is weakly disciplined, temporally unstable, and excessively resource-intensive.



(a) Discounted cost minimization only

(b) Dual-objective minimization

Figure 2: Dynamic optimal allocation of carbon credits to green projects

Consistent with the dual-target objective developed in Subection 2.2, we next activate the nominal policy resource commitment term in Eq. (7). This specification preserves the discounted policy preference expenditure term, but adds a second component of undiscounted nominal resource commitment that penalizes the total nominal expenditure of carbon credits and subsidies required to meet the same viability target. In the baseline analysis, we set μ=0.5, which assigns equal weights to the discounted expenditure objective and the nominal commitment objective. This neutral specification avoids imposing an ex-ante preference for either fiscal efficiency or resource preservation and therefore serves as a balanced benchmark. The sensitivity of the results to alternative weighting schemes is examined in Appendix A.5.

The resulting allocations are reported in Column 4 of Table 2 and illustrated in Fig. 2(b). Once the component of nominal resource commitment is introduced, the optimizer shifts credits from later operational years to the early stage of project operation, mainly 2017–2023. Because earlier credits carry higher present value, this front-loaded allocation offsets the same aggregate NPV deficit of 3337.04 million CNY with a substantially smaller physical volume of 54.23 million tons. These results highlight that the dual-target formulation is economically important as an allocation that is feasible under the weighted discounted policy-preference cost objective alone can still be highly inefficient in terms of physical resource use. The unconstrained dual-target solution therefore provides a natural benchmark for evaluating the constrained policy experiments that follow.

3.2.3. Scenario 3: Constrained optimal credit allocation

To mitigate greenhouse gas emissions, various regulatory and market-based mechanisms have been implemented globally (Zheng et al., 2021, 2022, 2023; Tan et al., 2023). The European Union introduced the ETS in 2005 (Ferrara and Giua, 2022), and China followed by launching pilot systems in 2011 to incentivize decarbonization (Chen et al., 2022). These mechanisms transform carbon emissions into a scarce resource, motivating enterprises to internalize their environmental impact.

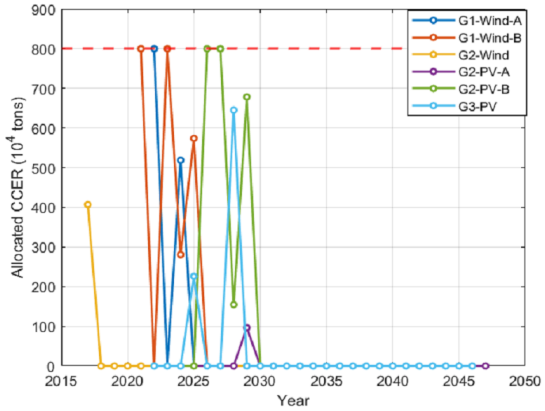
To reflect market-wide limits on carbon-credit supply, we next impose an annual aggregate cap on total credit allocation, $\sum_{i=1}^{I}\sum_{j=1}^{J} v^{b}_{i,j,l} \leq G_l$. This constraint, which captures aggregate scarcity in annual credit supply, is imposed at the system level and limits the total volume of credits that can be deployed across all projects in year l. While the aggregate discounted value needed to close the NPV deficit remains fixed at 3337.04 million CNY, the annual cap restricts when credits can be deployed and thereby reshapes the intertemporal allocation path.

The pattern shown in Fig. 3 is consistent with the dual-target mechanism developed above. In the unconstrained dual-target benchmark, the optimizer prefers to front-load credits because earlier credits have higher present value and therefore require smaller nominal quantities. Once an annual cap is imposed, however, the optimizer can no longer allocate all required credits in the earliest years. As G_l becomes tighter, a larger share of credit allocation must be shifted to later years of the project lifecycle. Since later credits are more heavily discounted, a larger nominal volume is needed to deliver the same present-value compensation.

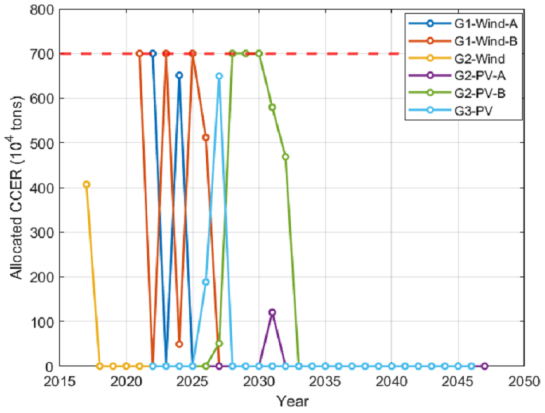
This mechanism explains the monotonic increase in total required credits across Columns 5–8 of Table 2. As the annual cap tightens from $G_l = 8$ million tons to $G_l = 5.5$ million tons, the total required allocation rises from 75.81 million tons to 145.34 million tons. The results demonstrate the capability of the proposed framework to quantify the resource cost of achieving financial viability under credit-supply constraints. While tighter annual caps can still achieve the same NPV-repair objective, they restrict the timing of credit deployment, force allocation into less efficient later periods, and increase the total nominal resource requirement.

3.2.4. Scenario 4: Synergistic optimization

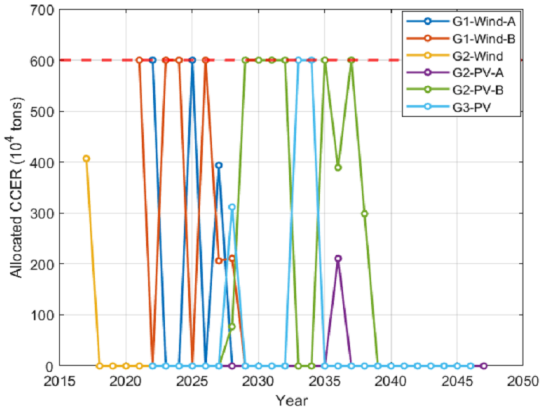
Renewable energy projects such as wind and PV have zero direct operational CO_2 emissions and therefore do not face compliance obligations under the national ETS. Instead, they participate in the carbon market through the CCER mechanism. As such, this scenario introduces a more realistic



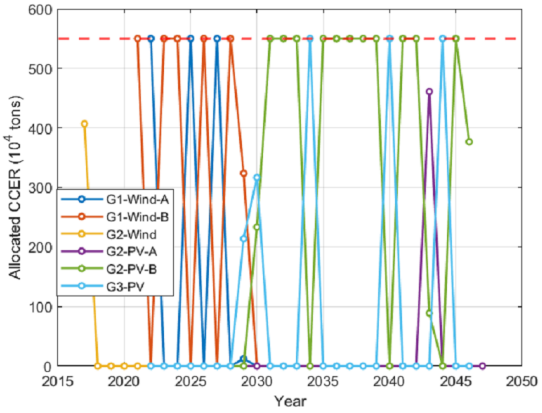
(a) $G_l = 8$ million tons (Total: 75.81 Mt)



(b) $G_l = 7$ million tons (Total: 85.76 Mt)



(c) $G_l = 6$ million tons (Total: 109.07 Mt)



(d) $G_l = 5.5$ million tons (Total: 145.34 Mt)

Figure 3: Dynamic optimal allocation of carbon credits with credit caps

institutional setting in which carbon-credit issuance is constrained by verified physical abatement, and subsidies are activated only when the carbon-market channel is insufficient. In our model, $v^b_{i,j,l}$ denotes the volume of CCERs authorized for certification for project (i, j) in year l. Project developers can generate additional revenue by selling these offsets to regulated emitters, who use them to satisfy their compliance obligations.[9]

Reflecting realistic institutional constraints, we impose a physical cap on credit issuance, where $v^b_{i,j,l} \leq a_{i,j,l}$. This constraint is compatible with, but structurally tighter than Eq. (11) ($v^b_{i,j,l} \leq v_{i,j,l}$). In conventional projects, credit allocations are capped by baseline emissions ($v_{i,j,l}$) to prevent over-allocation. By contrast, zero-emission renewable energy projects rely on counterfactual baseline emissions that are equal to the avoided emissions (i.e., $v_{i,j,l} = a_{i,j,l}$). Because CCER issuance is strictly limited to verified physical reductions, the constraint $v^b_{i,j,l} \leq a_{i,j,l}$ acts as a tighter, certified boundary specific to clean energy assets.

Table 3 reports the results of this physical issuance constraint. Columns 2–4 quantify the physical side of the carbon-market channel, including total CCER allocation, total abatement potential, and the utilization ratio. The remaining columns translate these physical allocations into financial terms by reporting the initial NPV deficit, the discounted value of CCER revenues, the residual deficit, and the percentage contribution of CCERs to deficit coverage. The table shows that the carbon-market channel alone is insufficient for most projects. Except for G3-PV, all projects exhaust their available CCER capacity but still retain positive residual deficits. For example, G2-PV-A obtains 92.05% deficit coverage from CCER revenues, whereas G2-Wind obtains only 13.51%. This heterogeneity reflects differences not only in the initial financing gap, but also in each project's verified abatement potential and the timing of available credit generation.

Table 3: Balance of CCER physical quantity and financial compensation with constraint ${v^b}_{i,j,l} \leq a_{i,j,l}$

Project	Total CCER (million tons)	Total Abate. a (million tons)	Utiliz. (%)	Initial Deficit (million CNY)	CCER NPV (million CNY)	Residual Deficit (million CNY)	Contribution (%)
G1-Wind-A	13.50	13.50	100	727.86	295.93	431.93	40.66
G1-Wind-B	8.04	8.04	100	1172.40	183.40	988.97	15.64
G2-Wind	1.52	1.52	100	248.07	33.51	214.56	13.51
G2-PV-A	1.35	1.35	100	30.84	28.39	2.45	92.05
G2-PV-B	22.56	22.56	100	850.17	510.35	339.83	60.03
G3-PV	7.40	20.94	35.34	307.70	307.70	0	100
Total	**54.37**	**67.91**	**80.06**	**3337.04**	**1359.28**	**1977.74**	**40.73**

Note: Utiliz. denotes the ratio of allocated CCERs to the physical emission reduction limit. A utilization rate of 100% accompanied by a positive residual deficit indicates that CCER revenues are insufficient to fully offset the project's financing gap. Contribution (%) is defined as CCER NPV (million CNY) divided by the initial deficit (million CNY). Minor discrepancies may arise due to rounding at different stages of computation. All values are reported to two decimal places for presentation purposes.

Fig. 4 shows how the binding physical-abatement constraint shapes the timing of credit deployment. Using G3-PV as an illustrative example, as G3-PV has sufficient abatement potential relative to its financing gap, the optimizer allocates CCERs up to the physical cap from 2022 to 2029 and only

[9]Aligned with the latest Administrative Measures for Voluntary Greenhouse Gas Emission Reduction Trading, `https://www.mee.gov.cn/xxgk2018/xxgk/xxgk02/202310/t20231020_1043694.html`

the remaining amount required in 2030, thereby fully closing the project's NPV deficit with a cumulative allocation of 7.40 million tons, no further credits are issued thereafter. In contrast, projects that already use 100% of their abatement potential but still exhibit residual deficits in Table 3 require an additional policy instrument.

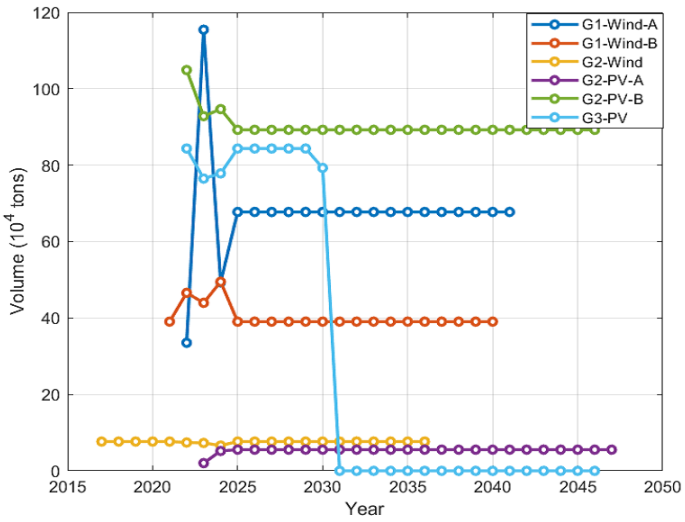


Figure 4: Dynamic allocation of CCERs under physical reduction constraints (${v^b}_{i,j,l} \leq a_{i,j,l}$)

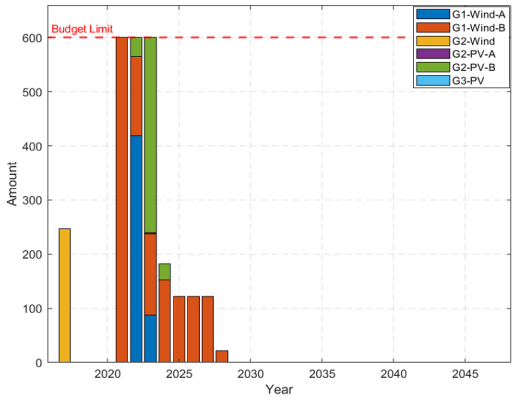
Thus, these residual deficits motivate the second stage of the synergistic policy design: direct subsidies are introduced as a fiscal backstop after the market-based CCER mechanism has been exhausted. To preserve fiscal discipline, we impose an annual subsidy budget constraint B_l as shown in Eq. (10). Table 4 reports the optimized subsidy requirements under alternative annual budget caps, while Fig. 5 shows the corresponding dynamic allocation paths. Columns 2–4 of Table 4 decompose each project's financing gap into the initial NPV deficit, the portion covered by CCER revenues, and the remaining NPV that must be covered by subsidies. The subsequent columns report the total nominal subsidy expenditure required when the annual budget cap is tightened from 600 million CNY to 310 million CNY. The caps $B_l = 600$, 500, and 400 million CNY represent progressively tighter fiscal environments, while $B_l = 310$ million CNY is the tightest feasible cap under the baseline carbon-price scenario. Below this level, the project-level break-even constraints cannot be satisfied given the residual deficits and the available CCER revenues. The $B_l = 310$ case therefore serves as a boundary stress test of the fiscal capacity required to maintain project-level solvency.

Specifically, under the tightest cap, $B_l = 310$ million CNY, total nominal subsidy expenditure rises to 4169.28 million CNY because the optimizer is forced to defer support into later, more heavily discounted periods, requiring larger nominal payments to deliver the same present-value compensation. This effect is concentrated in projects with large residual deficits. As G2-Wind enters operation relatively early and receives CCER support during the early allocation period, it faces less competition for scarce policy resources from other projects. Consequently, its subsidy allocation remains relatively stable across budget scenarios. By contrast, G1-Wind-B carries the largest residual financing gap and therefore exhibits a pronounced back-loaded subsidy profile, accounting for much of the

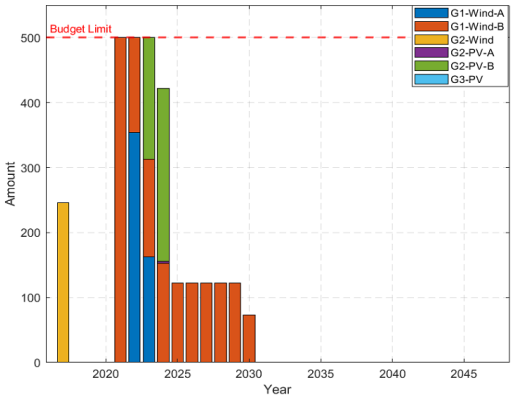
Table 4: Balance of CCER and subsidy with constraint ${v^b}_{i,j,l} \leq a_{i,j,l}$

Project	Initial Deficit (million CNY)	CCER NPV (million CNY)	Subsidy NPV (million CNY)	B_l=600 (million CNY)	B_l=500 (million CNY)	B_l=400 (million CNY)	B_l=310 (million CNY)
G1-Wind-A	727.86	295.93	431.93	507.22	516.92	553.77	594.16
G1-Wind-B	1172.40	183.40	988.97	1437.90	1634.40	1980.70	2681.00
G2-Wind	248.07	33.51	214.56	246.34	246.34	246.34	246.34
G2-PV-A	30.84	28.39	2.45	2.74	3.07	3.43	4.29
G2-PV-B	850.17	510.35	339.83	424.03	453.08	507.90	643.49
G3-PV	307.70	307.70	0	0	0	0	0
Total	**3337.04**	**1359.28**	**1977.74**	**2618.23**	**2853.81**	**3292.14**	**4169.28**

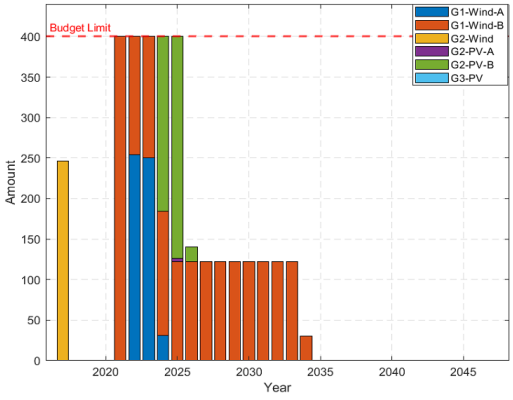
Note: Minor discrepancies may arise due to rounding at different stages of computation. All values are reported to two decimal places for presentation purposes.



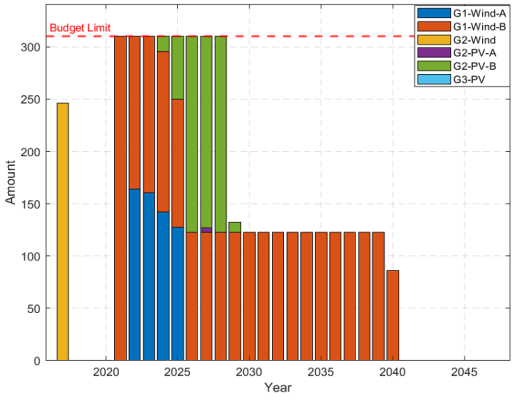
(a) B_l=600 million CNY (Total: 2618.23 million CNY)



(b) B_l=500 million CNY (Total: 2853.81 million CNY)



(c) B_l=400 million CNY (Total: 3292.14 million CNY)



(d) B_l=310 million CNY (Total: 4169.28 million CNY)

Figure 5: The optimal subsidy allocation under the annual total subsidy budget constraint

increase in total nominal subsidy expenditure under the tightest budget constraint.

Fig. 6 reveals the relationship between the subsidy coefficient, $k_{i,j,l}$, and the actual subsidy amount paid, $Subsidy_{i,j,l}$. Taking $B_l = 310$ million CNY as a representative case, the corresponding aggregate allocation results are summarized in Column 8 of Table 4, with a total nominal subsidy allocation of 4169.28 million CNY. Under this senario, a notable divergence, and in some instances an inverse relationship, exists between the subsidy coefficient and the actual cash flow allocated. For project G1-Wind-B, for example, the subsidy coefficient $k_{i,j,l} = 0.12$ in 2021 but reaches its upper bound ($k_{i,j,l} = 1$) in the subsequent years. However, the actual monetary subsidy paid in 2021 is the highest over the project lifecycle. This is primarily driven by the capital-intensive nature of green energy projects, where substantial upfront expenditures mean that even a modest coefficient can trigger significant fiscal outlays during the investment peak. This finding demonstrates that the subsidy coefficient should not be interpreted as a direct indicator of fiscal expenditure. Instead, actual subsidy payments depend jointly on the intervention intensity and the underlying investment scale. Regulators therefore need to calibrate both dimensions simultaneously to maintain fiscal efficiency and avoid unnecessary resource waste.

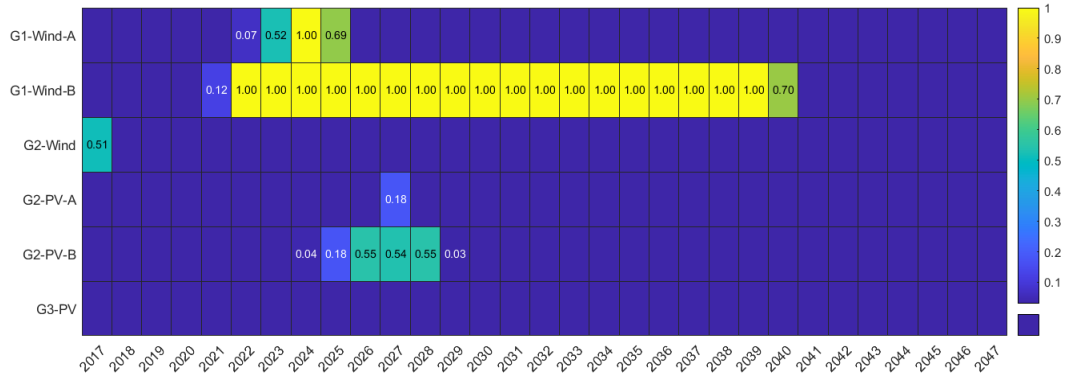
The project-level allocation patterns reveal why carbon credits and subsidies are complementary rather than interchangeable instruments. Dependence on policy intervention varies jointly with the size of the initial NPV deficit and each project's verified emission-reduction capacity. For example, although G1-Wind-B and G2-PV-B both face large initial deficits, the optimal policy mix differs substantially. G2-PV-B has sufficient abatement potential to cover a large share of its financing gap through CCER revenues, whereas G1-Wind-B's lower abatement capacity leaves a larger residual deficit that must be covered by subsidies. G3-PV represents the case in which the carbon-market channel alone restores financial viability. These contrasts show that the framework endogenously assigns market-based support to projects with stronger abatement capacity and reserves direct fiscal support for projects whose financing gaps cannot be closed through CCERs alone.

Overall, *Scenario 4* demonstrates the value of a hierarchical, synergistic intervention design. By first exhausting the CCER channel subject to the physical issuance constraint and then using subsidies only to cover residual deficits, the framework preserves the role of market-based incentives while providing a disciplined fiscal backstop. The combined policy mix restores project viability under strict institutional constraints without inefficiently crowding out the carbon-market mechanism. Relative to a single-instrument design, it therefore provides a more flexible and resource-efficient approach to aligning environmental performance with financial feasibility.

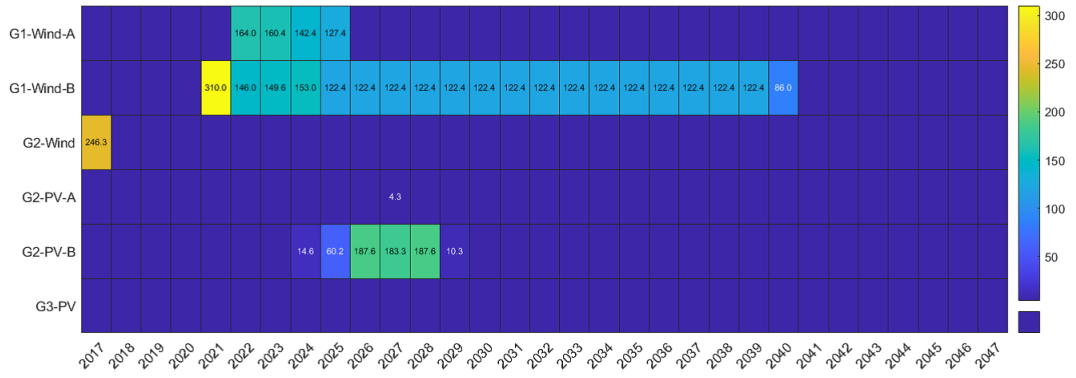
4. Green bond pricing analysis

4.1. Valuation of the embedded green option

Following the optimization analysis, the second stage of our framework translates the optimized policy interventions into green bond pricing implications. In doing so, we incorporate the policy-adjusted project cash flows into the option-pricing model and valuing the resulting embedded green attribute. The pricing exercise is conducted for the four policy states generated in the preceding analysis: the no-intervention baseline case, the sufficient-credit case, the physically constrained credit-allocation case, and the synergistic credit-subsidy case. The four cases differ in how policy support



(a) Subsidy coefficient $k_{i,j,l}$



(b) Actual subsidy amount $Subsidy_{i,j,l}$

Figure 6: Detailed subsidy allocation under the annual total subsidy budget constraint

Note: Comparison of subsidy coefficients in panel (a) and actual monetary allocations in panel (b) for constraint $B_l = 310$ million CNY. The corresponding aggregate allocation results over the project lifecycle are reported in Column 8 of Table 4. Specifically, the CCER NPV is 1359.28 million CNY, the subsidy NPV is 1977.74 million CNY, and the nominal subsidy allocation amounts to 4169.28 million CNY.

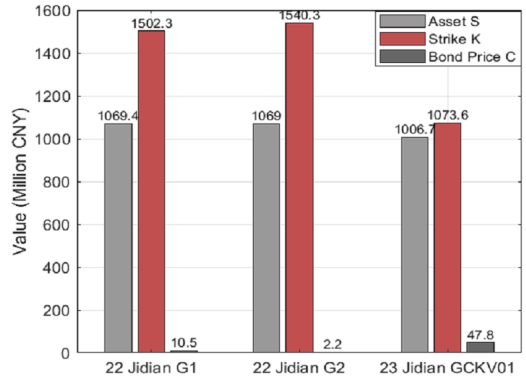
changes the option-pricing inputs. Credit-based intervention primarily raises the revenue-side asset value S, while subsidy-based intervention reduces the cost-side strike value K after available CCERs have been exhausted. This distinction allows the pricing model to separate revenue-side and cost-side channels, and to evaluate how alternative policy instruments repair project cash flows and convert environmental performance into embedded green option value.

Fig. 7 reports how the embedded green option responds to alternative policy-intervention states. Panel (a) illustrates the baseline scenario without intervention. For all bonds, the present value of costs (K) exceeds that of revenues (S). From a project perspective, the negative NPV renders the underlying projects economically unviable. From an option pricing perspective, the strike price (K) sits far above the asset price (S), placing the embedded green option in a deeply OTM state. The option value C remains positive but small, reflecting only residual time value that there is some probability that asset volatility, σ, raises future revenues above costs before maturity. Economically, however, this residual option value is not sufficient to compensate investors for the underlying cash-flow shortfall. The baseline case therefore illustrates why, absent policy repair, the green attribute remains latent and the bonds are difficult to support on project cash flows alone.

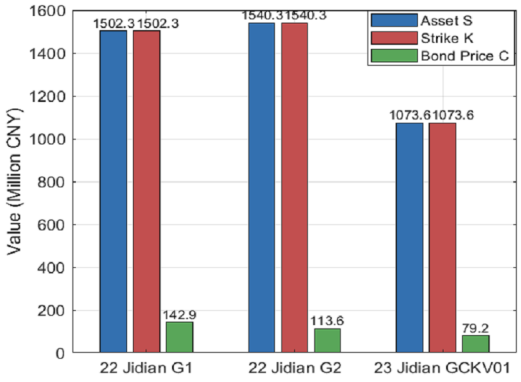
Panel (b) of Fig. 7 presents the pricing outcomes under the optimized credit scenario. The optimization stage closes the project-level financing gap by raising the revenue-side asset value S until it matches K. This policy intervention shifts the embedded option from an OTM state to an ATM state. The resulting option value C becomes economically meaningful because, once the downside cash-flow gap is repaired, investors can value the upside potential generated by project volatility. Furthermore, although all bonds satisfy $S = K$, their green option prices differ. In particular, 22 Jidian G1 > 22 Jidian G2 > 23 Jidian GCKV01, which reflects sensitivity of option value to bond volatility and maturity. For instance, despite 22 Jidian G1 having lower revenue and cost levels than 22 Jidian G2, its valuation is higher due to its longer maturity, which amplifies the time value of the embedded option.

Panel (c) of Fig. 7 introduces the institutional constraint $v^b_{i,j,l} \leq a_{i,j,l}$. Here, even after available CCERs are exhausted, the revenues of 22 Jidian G1 and 22 Jidian G2 remain below their costs (i.e., $S < K$), so their embedded green options remain in an OTM state. Their valuations improve relative to the no-intervention baseline because CCER revenues partially reduce the financing gap, but the projects are not fully repaired. In contrast, 23 Jidian GCKV01 has sufficient abatement capacity relative to its deficit and reaches the ATM state, thereby unlocking its embedded option value. Panel (c) therefore shows that carbon-market support can generate option value only to the extent that verified physical abatement is large enough to close the financial gap.

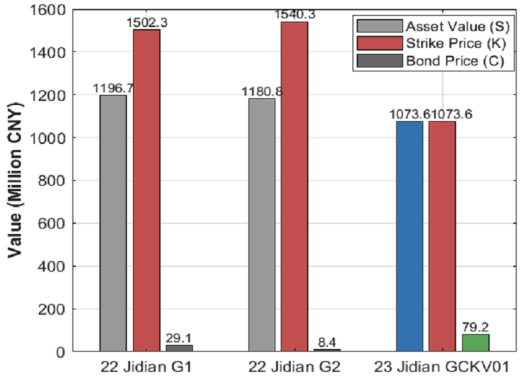
Panel (d) of Fig. 7 reports the pricing dynamics of the synergistic credit-subsidy case, in which subsidies are introduced after the available CCER channel has been exhausted. The subsidy backstop closes the remaining financing gaps by reducing the effective cost-side strike value K, while CCERs continue to operate through the revenue side. Although both Panel (b) and Panel (d) achieve the ATM state, they do so through different cash-flow channels. Panel (b) reflects a revenue-side repair achieved through sufficient carbon-credit allocation, whereas Panel (d) reflects a combined repair through carbon-market revenues and cost-side subsidy support. The critical economic distinction lies in the endogenous adjustment of the cost structure induced by the subsidy channel. A credit-



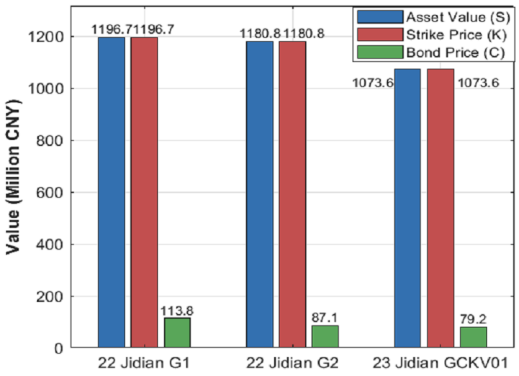
(a) Baseline: Distressed OTM State



(b) Optimized credits: Repaired ATM State



(c) Insufficient credits: Distressed OTM State



(d) Synergistic policy: Repaired ATM State

Figure 7: Comparison of pricing results in different scenarios

Note: Panel (a) reports the no-intervention baseline, in which project revenues are insufficient to cover costs and the embedded green option remains in a distressed out-of-the-money (OTM) state. Panel (b) reports the optimized-credit case, where optimized CCER allocation fully repairs the project cash flows and moves the option to a repaired at-the-money (ATM) state. Panel (c) imposes the physical reduction constraint under which available CCERs are insufficient for some projects, so the corresponding options remain distressed. Panel (d) reports the synergistic policy case, where subsidies are introduced after available CCERs are exhausted, restoring financial viability and moving the embedded option back to a repaired ATM state.

based repair raises the revenue-side asset value S, whereas a subsidy-based repair lowers the effective cost-side strike value K. Because option values are nonlinear in S, K, maturity, and volatility, these two policy channels can generate different values of C even when both restore the same break-even condition, $S = K$. The resulting differences in show that green bond valuation depends not only on the elimination of the NPV shortfall, but also on the incidence, timing, and composition of the policy-induced cash-flow adjustment.

The results in Fig. 7 demonstrates the pricing implications of the proposed two-stage framework. The optimized policy intervention improves the risk-return characteristics of green assets by shifting the embedded option from a distressed OTM state to a repaired ATM state. Once the downside cash-flow gap is closed, project volatility becomes valuable rather than purely risky. The option value C therefore measures the financial value of the embedded green attribute that is activated by policy support. This option-based framework goes beyond traditional, static break-even logic, providing a structural interpretation of the greenium. Policy support internalizes environmental value into project cash flows and allows the market to price the upside optionality of green assets under uncertainty.

4.2. Structural explanation of the greenium

The preceding analysis values the embedded green option, C, created by policy-supported project repair. We now examine the endogenous pricing structure in the primary bond market, providing a benchmark for how much green value could be reflected in bond pricing if the policy-induced environmental option were fully capitalized by investors. In particular, we decompose the total value of green bonds into a conventional fixed-income component (P_{fixed}) and an embedded green option component (C):

$$P_{GB} = P_{fixed} + C, \tag{16}$$

where

$$P_{fixed} = \sum_{t=1}^{T} \frac{Principal \times Coupon}{(1 + r_b)^t} + \frac{Principal}{(1 + r_b)^T}. \tag{17}$$

The benchmark discount rate, r_b, is defined as the yield on a matched conventional bond with similar credit risk and maturity. Following a matching approach that controls for market-wide interest rate conditions and issuer-specific characteristics, we use 23 Jidian MTN001 as the reference bond, and its coupon rate is set as the benchmark discount rate, i.e., $r_b = 3.28\%$. This approachcompares green bonds with counterfactual conventional bonds to estimate the greenium (Zerbib, 2019).

The resulting decomposition of P_{GB} is reported in Table 5, which is based on pricing Scenario 4: Synergistic policy in Fig. 7(d) and corresponds to the most stringent constraint setting. The table shows that P_{GB} of each green bond exceeds both its par value and the conventional fixed-income component once the embedded green option is included. Structurally, the fixed-income component, P_{fixed}, captures the present value of promised coupon and principal payments under the benchmark yield, while the option component, C, reflects the incremental value generated by policy-supported improvements in project cash-flow structure. This decomposition provides a structural interpretation of the greenium as an option-like payoff induced by environmental policy support, rather than a premium mechanically attached to the green label.

Table 5: Valuation components of Jidian green bonds

Bond Name	Principal (million CNY)	Maturity (Years)	Coupon Rate	Fixed Value P_{fixed} (million CNY)	Green Option C (million CNY)	Total Value P_{GB} (million CNY)
22 Jidian G1	1000	5	3.80%	1023.69	113.85	1137.54
22 Jidian G2	1000	3	2.90%	989.26	87.09	1076.35
23 Jidian GCKV01	500	3	2.99%	495.92	79.19	575.11

Note: The resulting decomposition of P_{GB} is based on pricing Scenario 4: Synergistic policy in Fig. 7 (d) and corresponds to the most stringent constraint setting.

To further quantify the structural valuation in yield-equivalent terms, we define the implied shadow yield, y_g, as the discount rate that equates the contractual cash flows of the green bond to its total value, P_{GB}. The implied shadow yield can be obtained from solving the following equation:

$$P_{GB}=\sum_{t=1}^{T}\frac{Principal\times Coupon}{(1+y_g)^t}+\frac{Principal}{(1+y_g)^T}. \tag{18}$$

This transformation allows us to compare three yield concepts. The benchmark yield, r_b, represents the fair financing cost of comparable conventional debt and captures the issuer's credit risk absent green attributes. The contractual coupon, r_c, is the actual coupon paid by the issuer in the primary market. The implied shadow yield, y_g, is the theoretical yield that would prevail if the embedded green option value were fully internalized into the bond price. Based on these three yield measures, we construct three greenium metrics. The observed greenium, $r_b - r_c$, measures the realized coupon concession relative to the matched conventional benchmark and captures the portion of green value already reflected in primary-market issuance pricing. The potential greenium, $r_b - y_g$, measures the model-implied financing-cost reduction that would prevail if the embedded green option were fully capitalized into the bond price. The greenium gap, $r_c - y_g$, measures the uncapitalized component of green value, or equivalently the wedge between the issuer's actual financing cost and the shadow financing cost implied by the structural value of the green attribute. Table 6 reports the results.

Table 6: Benchmark, contractual, and shadow yields with implied greenium measures for Jidian green bonds

Bond Name	Benchmark Yield (r_b)	Coupon (r_c)	Shadow Yield (y_g)	Observed Greenium ($r_b - r_c$)	Potential Greenium ($r_b - y_g$)	Greenium Gap ($r_c - y_g$)
22 Jidian G1	3.28%	3.80%	0.97%	-52 bps	231.13 bps	283.13 bps
22 Jidian G2	3.28%	2.90%	0.34%	38 bps	294.22 bps	256.22 bps
23 Jidian GCKV01	3.28%	2.99%	-1.83%	29 bps	511.47 bps	482.47 bps

Note: The observed greenium measures the coupon saving relative to the matched conventional benchmark. The potential greenium measures the counterfactual financing-cost reduction if the embedded green option were fully internalized into the bond price. The greenium gap is the difference between the contractual coupon and the implied shadow yield.

The comparison in Table 6 reveals a substantial gap between the green value implied by the structural model and the green value reflected in observed issuance coupons. For 22 Jidian G1, the observed greenium is negative, that is, its coupon exceeds the matched conventional benchmark by 52 bps. This suggests that, at the time of issuance, investors did not price the bond as receiving a

financing benefit from its green attribute, possibly because project risk and market uncertainty dominated the environmental value channel. By contrast, 22 Jidian G2 and 23 Jidian GCKV01 exhibit positive observed greenium of 38 bps and 29 bps, respectively, indicating some coupon advantage relative to the conventional benchmark. However, note that these observed greenium remain far below the corresponding potential greenium implied by the structural valuation.

The greenium gap therefore captures the degree of incomplete internalization of environmental value in primary-market pricing. For 22 Jidian G2, the potential greenium is 294.22 bps, while the observed greenium is only 38 bps, leaving a gap of 256.22 bps. For 23 Jidian GCKV01, the gap is even larger, at 482.47 bps. Here, y_g should be interpreted as a shadow price that summarizes the financing-cost equivalent of the embedded environmental option under full internalization. The negative shadow yields arise because the option-augmented structural value of the bond, $P_{GB} = P_{fixed} + C$, is sufficiently high relative to its promised cash flows.

Economically, the results indicate that current green bond coupons internalize only part of the environmental option value created by policy-supported project repair. Several frictions can explain this wedge, including limited pledgeability of future environmental benefits, uncertainty about the persistence of policy support, verification and certification risks, and the time lag between current emission reductions and long-term environmental gains. The structural model makes this wedge explicit by distinguishing the green value created at the project level from the portion capitalized in the issuance coupon.

This analysis provides a structural explanation for the greenium. The greenium is not treated as a reduced-form pricing anomaly or a label-based discount. Instead, it arises from a repaired asset structure in which policy intervention internalizes environmental value into project cash flows and converts that value into an embedded option component. The observed greenium measures the part of this value already reflected in market pricing, while the greenium gap measures the remaining latent value that could be capitalized through stronger policy credibility, better verification, or financial structures that make environmental cash flows more pledgeable.

5. Extensions: Robustness checks

5.1. With a weaker viability target ($NPV_{i,j} \geq 0$)

The baseline model assumes a strict project-level break-even target, $NPV_{i,j} = 0$. This equality target provides a transparent benchmark for policy repair, but in practice regulators may not be able to calibrate carbon-credit and subsidy interventions with exact precision. Therefore, in this subsection, we conduct a robustness check by examining a weaker viability requirement, $NPV_{i,j} \geq 0$, under which the regulator only requires each project to be financially viable and does not explicitly prohibit positive surplus. This specification tests whether the main results are driven by the equality constraint itself or by the cost-minimizing structure of the policy problem.

Table 7 reports the relaxed viability results under the same boundary fiscal environment used in the synergistic benchmark, namely the $B_l = 310$ million CNY column of Table 4. The relaxed inequality constraint produces the same allocation as the baseline case. This consistency is driven by an economic outcome rather than a mechanical restriction. Since both carbon credits and subsidies

are costly policy instruments, any allocation with $NPV_{i,j} > 0$ would represent overcompensation and inefficient use of scarce policy resources. The optimizer therefore reduces support until the viability constraint binds. As a result, the cost-minimizing solution under $NPV_{i,j} \geq 0$ endogenously converges to the same marginal break-even outcome, $NPV_{i,j} = 0$, for each project.

Table 7: Intervention outcomes under the relaxed viability constraint ($NPV_{i,j} \geq 0$)

Project	Initial Deficit (million CNY)	CCER NPV (million CNY)	Subsidy NPV (million CNY)	Surplus NPV (million CNY)	B_l=310 (million CNY)
G1-Wind-A	727.86	295.93	431.93	0	594.16
G1-Wind-B	1172.40	183.40	988.97	0	2681.00
G2-Wind	248.07	33.51	214.56	0	246.34
G2-PV-A	30.84	28.39	2.45	0	4.29
G2-PV-B	850.17	510.35	339.83	0	643.49
G3-PV	307.70	307.70	0	0	0
Total	**3337.04**	**1359.28**	**1977.74**	**0**	**4169.28**

Note: The results are computed under the same annual subsidy budget constraint as the project-level synergistic benchmark in Table 4, namely the boundary case $B_l = 310$ million CNY. This value represents the tightest feasible annual subsidy cap under the baseline carbon-price scenario. Minor discrepancies may arise due to rounding at different stages of computation. All values are reported to two decimal places for presentation purposes.

This result confirms the robustness of the just-enough intervention principle. Even when the regulator allows positive project surplus, the optimal policy does not over-support green projects because any excess support increases the objective value without relaxing a binding constraint. The project-level break-even condition in the baseline model should therefore be interpreted not only as a modelling assumption, but also as the equilibrium outcome of cost-efficient policy design. In practice, the $NPV_{i,j} = 0$ condition is consistent with competitive procurement mechanisms, such as reverse auctions and bidding schemes in China's renewable energy sector, in which developers reveal cost structures and policy support is disciplined toward marginal financial viability. Accordingly, the main conclusions on green asset repair, embedded option valuation, and the resulting greenium remain valid under the weaker viability target.

5.2. Aggregate efficiency versus bond-level solvency

We next examine how the boundary of the viability constraint affects the trade-off between fiscal efficiency and market investability. In doing so, instead of requiring every project to satisfy the project-level break-even condition, we first impose an aggregate portfolio-level viability constraint, $\sum_{i=1}^{I} \sum_{j=1}^{J} NPV_{i,j} = 0$. This specification gives the regulator maximum flexibility in allocating policy resources across the entire green investment portfolio. It allows surplus generated by more policy-efficient projects, such as projects with larger abatement potential or more favorable cash-flow profiles, to offset the deficits of weaker projects. The potential benefit is lower total intervention cost, while the potential cost is that individual projects or bond pools may remain financially impaired even when the aggregate portfolio breaks even.

Table 8 reports the optimized intervention outcomes under this aggregate constraint. Relative to the project-level benchmark, the portfolio-level constraint substantially reduces the total nominal subsidy requirement from 4169.28 million CNY (see the Column 8 of Table 4) to 2416.38 million CNY (Column 6 of Table 8). This fiscal saving arises because the optimizer reallocates support toward

projects with the highest policy efficiency. As inllustrated in Fig. 8, G3-PV receives CCER support beyond its own break-even requirement and generates a positive NPV surplus, which is then used at the portfolio level to offset deficits elsewhere. The aggregate constraint therefore improves fiscal efficiency by allowing cross-subsidization across projects.

Table 8: Detailed intervention results under the aggregate portfolio-level constraint ($\sum_{i=1}^{I}\sum_{j=1}^{J} NPV_{i,j} = 0$)

Project	Initial Deficit (million CNY)	CCER NPV (million CNY)	Subsidy NPV (million CNY)	Net NPV Surplus (million CNY)	Total Nominal Subsidy (million CNY)
G1-Wind-A	727.86	295.93	0	-431.93	0
G1-Wind-B	1172.40	183.40	270.01	-718.96	310.00
G2-Wind	248.07	33.51	313.99	99.43	376.14
G2-PV-A	30.84	28.39	259.34	256.89	322.54
G2-PV-B	850.17	510.35	0	-339.83	0
G3-PV	307.70	460.96	981.13	1134.40	1407.70
Total	**3337.04**	**1512.54**	**1824.47**	**0**	**2416.38**

Note: The results are computed under the same annual subsidy budget constraint as the project-level synergistic benchmark in Table 4, namely the boundary case $B_l = 310$ million CNY. This value represents the tightest feasible annual subsidy cap under the baseline carbon-price scenario. Minor discrepancies may arise due to rounding at different stages of computation. All values are reported to two decimal places for presentation purposes.

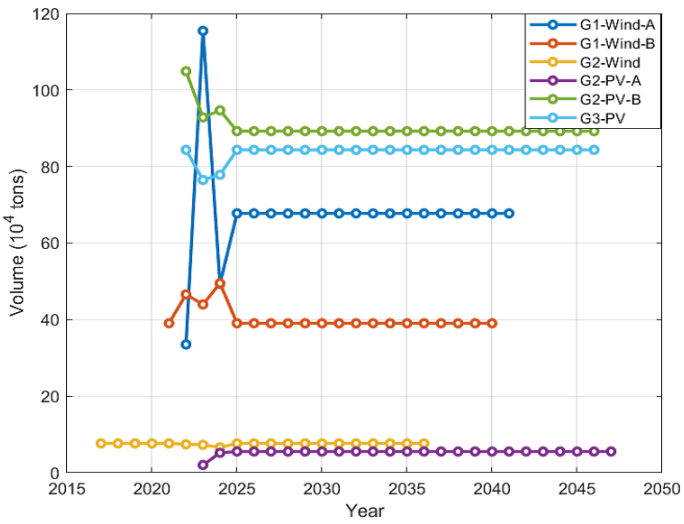


Figure 8: Annual CCER allocation under the aggregated portfolio-level constraint ($\sum_{i=1}^{I}\sum_{j=1}^{J} NPV_{i,j} = 0$)

This aggregate efficiency gain, however, comes at the cost of bond-level solvency. Fig. 9 shows that subsidies become concentrated in a subset of projects, especially G3-PV, which even did not exhaust its physical caps in the baseline scenario. This concentration is optimal for portfolio-level cost minimization, but it allows some projects and project pools to remain below break-even. The pricing result is shown in Fig. 10. Under the aggregate portfolio-level constraint, 22 Jidian G1 fails to cover its costs, leaving its embedded green option in a distressed OTM state. This outcome highlights a critical misalignment between macro-policy objectives and the financing vehicle: if surplus from one bond-financed project pool is allocated to support unrelated assets, the investability of individual bonds can be undermined despite achieving aggregate solvency.

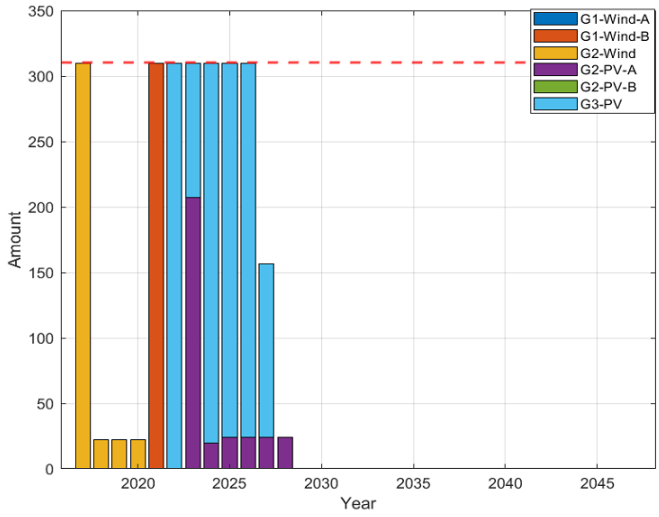


Figure 9: Subsidies allocation under the aggregated portfolio-level constraint ($\sum_{i=1}^{I}\sum_{j=1}^{J} NPV_{i,j} = 0$)

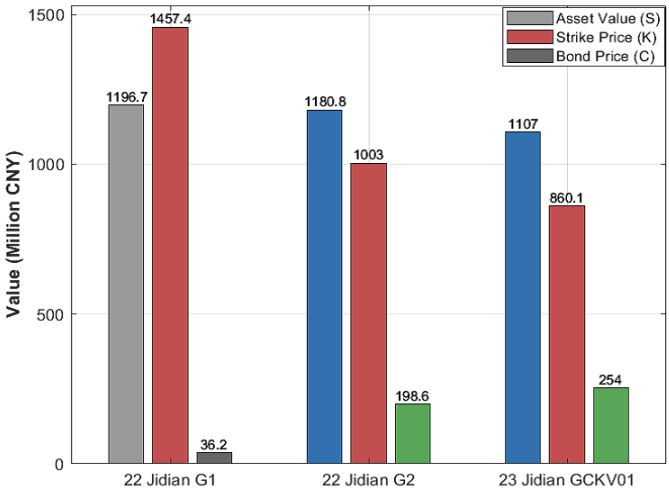


Figure 10: Embedded green option valuation under the portfolio-level constraint ($\sum_{i=1}^{I}\sum_{j=1}^{J} NPV_{i,j} = 0$)

Therefore, we consider a second relaxation that aligns the intervention boundary with the financial instrument rather than with either individual projects or the full portfolio. Specifically, we impose a bond-level viability constraint, $\sum_{j \in J_i} NPV_{i,j} = 0$ for each bond i, where J_i denotes the set of projects financed by bond i. This approach allows cross-subsidization within each bond-financed project pool, while preventing cash-flow support from being transferred across unrelated bonds. This boundary matches the credit-relevant unit faced by investors because bondholders are ultimately exposed to the aggregate cash-flow capacity of the project pool financed by each bond rather than to the standalone viability of every underlying project.

Table 9 reports the bond-level optimization results. The total nominal subsidy requirement is 3085.66 million CNY, which lies between the project-level benchmark of 4169.28 million CNY (see Table 4) and the aggregate portfolio-level result of 2416.38 million CNY (see Table 8). As a result, bond-level targeting preserves part of the fiscal efficiency gained from internal cross-subsidization, while avoiding the solvency failures created by unrestricted portfolio-level pooling. The allocation of CCERs remains identical to that reported in Fig. 4, except for G3-PV where carbon-credits are not fully exhausted, all other projects fully utilize their available credits prior to the activation of fiscal subsidies. Then, Fig. 11 illustrates how the optimizer redistributes subsidies within each bond pool to satisfy the bond-level break-even condition.

Table 9: Intervention results under bond-level constraints ($\sum_{j \in J_i} NPV_{i,j} = 0$)

Project/Bond	Initial Deficit (million CNY)	CCER NPV (million CNY)	Subsidy NPV (million CNY)	Net NPV Surplus (million CNY)	Total Nominal Subsidy (million CNY)
G1-Wind-A	727.86	295.93	1005.00	573.12	1677.00
G1-Wind-B	1172.40	183.40	415.86	-573.12	581.88
22 Jidian G1	1900.26	479.33	1420.86	0	2258.88
G2-Wind	248.07	33.51	313.99	99.43	376.14
G2-PV-A	30.84	28.39	97.59	95.14	165.23
G2-PV-B	850.17	510.35	145.26	-194.57	285.41
22 Jidian G2	1129.08	572.25	556.84	0	826.78
G3-PV	307.70	307.70	0	0	0
23 Jidian GCKV01	307.70	307.70	0	0	0
Total	**3337.04**	**1359.28**	**1977.70**	**0**	**3085.66**

Note: The results are computed under the same annual subsidy budget constraint as the project-level synergistic benchmark in Table 4, namely the boundary case $B_l = 310$ million CNY. This value represents the tightest feasible annual subsidy cap under the baseline carbon-price scenario. Minor discrepancies may arise due to rounding at different stages of computation. All values are reported to two decimal places for presentation purposes.

The pricing results under bond-level optimization are shown in Fig. 12. Unlike the aggregate portfolio-level constraint, the bond-level constraint restores the financial viability of each bond-financed pool. The asset values (S) of 22 Jidian G1 and 22 Jidian G2 slightly exceed their cost (K), while 23 Jidian GCKV01 remains stable as it is already fully supported by CCER revenues. This is because project-level surpluses can offset project-level deficits within the same financing vehicle, creating a buffer for bondholders without diverting resources across unrelated bonds. It reflects the diversification benefit within each bond pool and indicates that structural diversity in the financed project portfolio can enhance risk resilience. Bond-level targeting therefore avoids the sacrifice effect observed under portfolio-level optimization and preserves the embedded green option value at the instrument level. Consequently, even when individual projects exhibit moderate NPV dispersion, the

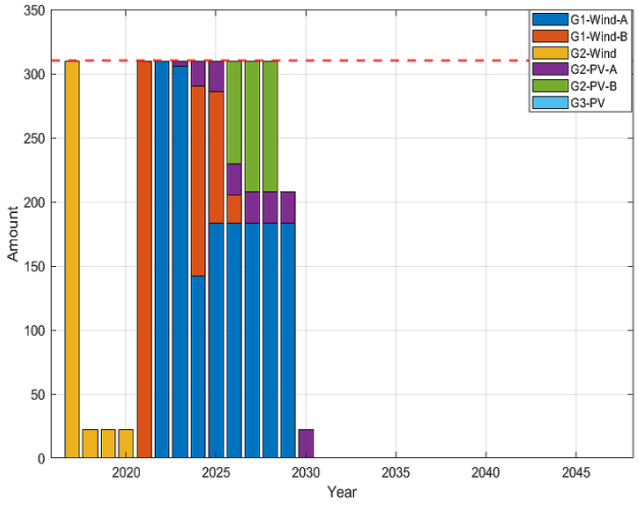


Figure 11: Subsidy allocation under the bond-level constraint ($\sum_{j \in J_i} NPV_{i,j} = 0$)

aggregate revenue of the bond-financed pool remains sufficient to support robust bond pricing.

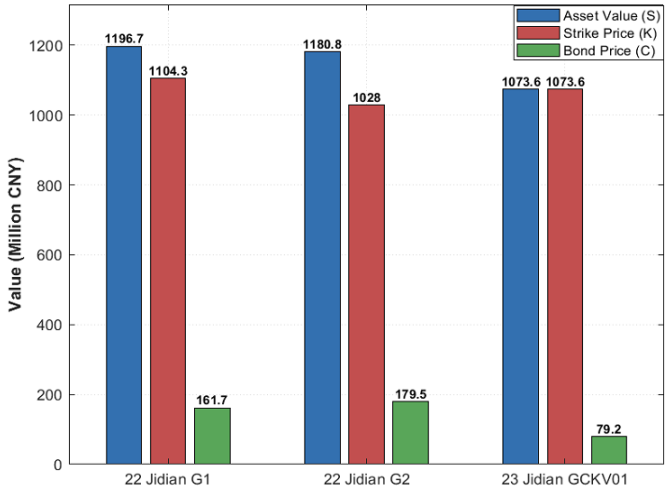


Figure 12: Embedded green option valuation under the bond-level constraint ($\sum_{j \in J_i} NPV_{i,j} = 0$)

Overall, the comparison between portfolio-level and bond-level optimization shows that the relevant boundary of policy intervention matters for green bond pricing. Project-level targeting protects every asset but requires the largest nominal subsidy commitment. Aggregate portfolio targeting minimizes fiscal outlays but may leave individual bonds financially impaired. Bond-level targeting preserves fiscal efficiency through within-bond pooling while maintaining the solvency of the financial instruments held by investors, and hence, offers an intermediate and economically practical design. This alignment between the policy boundary and the financing boundary is essential for preventing policy-induced misallocation and sustaining investor confidence in the green bond market.

6. Extensions: Dynamic Analysis

To evaluate the robustness of the proposed framework under carbon market volatility, we extend the analysis by allowing the carbon price, p_l^c, to evolve over time. This extension addresses a critical policy concern: how should the mix of CCER allocation and fiscal subsidies adjust when carbon prices deviate from the baseline path? We consider two symmetric carbon price trajectories, an appreciation scnario of +5% per year and a depreciation scenario of −5% per year. These scenarios allow us to examine whether the optimized policy mix can be dynamically recalibrated to maintain project solvency under both favorable and adverse carbon market conditions.

6.1. The appreciation scenario

The optimization results for the appreciation scenario are presented in Table 10. Compared with the baseline results in Table 4, a rising carbon price increases the present value of CCER revenues and reduces the need for direct fiscal support. Most notably, for project G2-PV-A, the appreciation in CCER value allows the project to achieve break-even status without government subsidies. At the portfolio level, the contribution of CCER revenues increases from 40.73% in the the baseline scenario to 52.28%, while total nominal subsidies decline to 2946.59 million CNY. This result shows that a stronger carbon market can operate as a fiscal hedge, shifting part of the intervention burden from public subsidies to market-based carbon revenues.

Table 10: Intervention results under dynamic carbon price appreciation (+5% p.a.)

Project	Target Deficit (million CNY)	CCER NPV (million CNY)	Subsidy NPV (million CNY)	Contribution (%)	CCER Qty (million tons)	Nominal Subsidy (million CNY)
G1-Wind-A	727.86	398.75	329.11	54.78	13.50	422.13
G1-Wind-B	1172.40	231.93	940.44	19.78	8.04	2115.30
G2-Wind	248.07	35.21	212.86	14.20	1.52	244.39
G2-PV-A	30.84	30.84	0	100	0.68	0
G2-PV-B	850.17	740.34	109.83	87.08	22.56	164.77
G3-PV	307.70	307.70	0	100	5.95	0
Total	**3337.04**	**1744.77**	**1592.24**	**52.28**	**52.25**	**2946.59**

Note: Contribution refers to the proportion of the initial deficit covered by CCER revenue. The results are computed under the same annual subsidy budget constraint as the project-level synergistic benchmark in Table 4, namely the boundary case $B_l = 310$ million CNY. This value represents the tightest feasible annual subsidy cap under the baseline carbon-price scenario. Minor discrepancies may arise due to rounding at different stages of computation. All values are reported to two decimal places for presentation purposes.

Fig. 13 further shows how carbon price appreciation changes the timing of CCER allocation. Specifically, the CCER allocations for G3-PV and G2-PV-A reach the break-even threshold earlier than in the baseline. This demonstrates that, once credit allocation renders these projects financially viable, the flexibility is created to reallocate subsidies towards more financially distressed projects, as shown in Fig. 14.

6.2. The depreciation scenario

We next examine a carbon price depreciation scenario of −5% per year to assess policy resilience under adverse market conditions. As shown in Table 11, a persistent decline in carbon prices weakens the revenue-generating capacity of CCERs, reducing the NPV and relative contribution of carbon

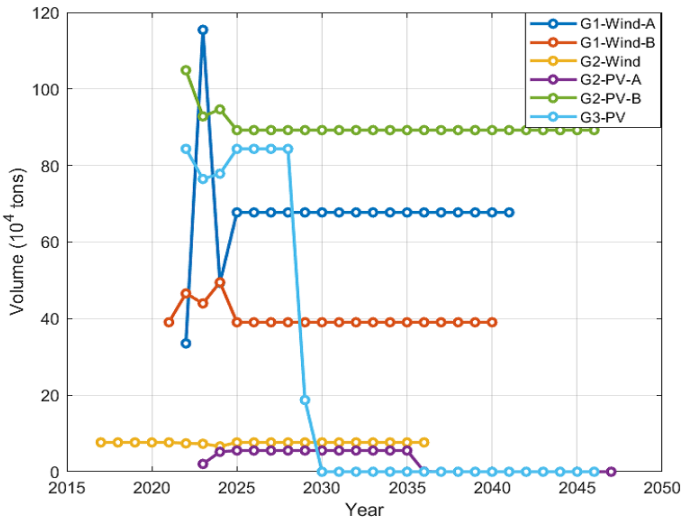


Figure 13: Dynamic optimal allocation of CCERs under the carbon price appreciation scenario

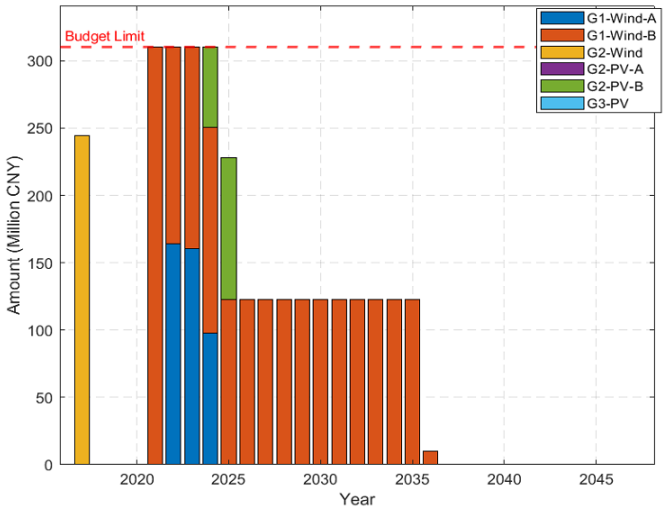


Figure 14: Dynamic optimal allocation of subsidies under the carbon price appreciation scenario

assets across all projects. The present value of CCER revenues falls to 1121.23 million CNY, and their contribution declines to 33.60%. As a result, the subsidy NPV increases to 2215.83 million CNY, indicating a larger fiscal burden on the government. Fig. 15 shows the corresponding dynamic CCER allocation. Compared with the baseline allocation in Fig. 4, the most pronounced adjustment occurs for project G3-PV, whose break-even point is delayed from 2030 to 2035.

Table 11: Intervention results under dynamic carbon price depreciation (−5% p.a.)

Project	Target Deficit (million CNY)	CCER NPV (million CNY)	Subsidy NPV (million CNY)	Contribution (%)	CCER Qty (million tons)	Nominal Subsidy (million CNY)
G1-Wind-A	727.86	229.48	498.38	31.53	13.50	676.37
G1-Wind-B	1172.40	151.92	1020.50	12.96	8.04	2667.80
G2-Wind	248.07	33.81	214.26	13.63	1.52	246.00
G2-PV-A	30.84	19.26	11.58	62.45	1.35	22.65
G2-PV-B	850.17	379.06	471.11	44.59	22.56	884.25
G3-PV	307.70	307.70	0	100	11.63	0
Total	**3337.04**	**1121.23**	**2215.83**	**33.60**	**58.60**	**4497.07**

Note: Contribution refers to the proportion of the initial deficit covered by CCER revenue. The boundary benchmark in Table 4 uses $B_l = 310$ million CNY, which is the tightest feasible annual subsidy cap under the baseline carbon-price scenario. Under carbon price depreciation, this cap is no longer feasible because lower CCER values increase the residual financing gap. The results therefore use a relaxed annual subsidy cap of $B_l = 350$ million CNY to restore project-level solvency. Minor discrepancies may arise due to rounding at different stages of computation. All values are reported to two decimal places for presentation purposes.

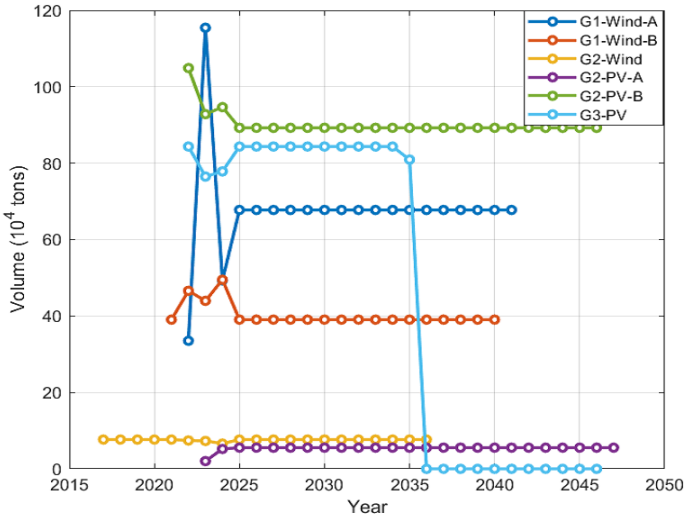


Figure 15: Dynamic optimal allocation of CCERs under the carbon price depreciation scenario

The depreciation scenario also shows that fiscal rules must remain sufficiently flexible when carbon market conditions deteriorate. Under the original annual subsidy cap of $B_l = 310$ million CNY, the policy mix cannot restore all projects to the break-even threshold because the decline in carbon prices erodes the value of CCER revenues. We therefore relax the annual subsidy cap to $B_l = 350$ million CNY. The resulting subsidy allocation, shown in Fig. 16, illustrates the additional fiscal support required to offset the loss of carbon-market value and maintain project solvency during a market downturn.

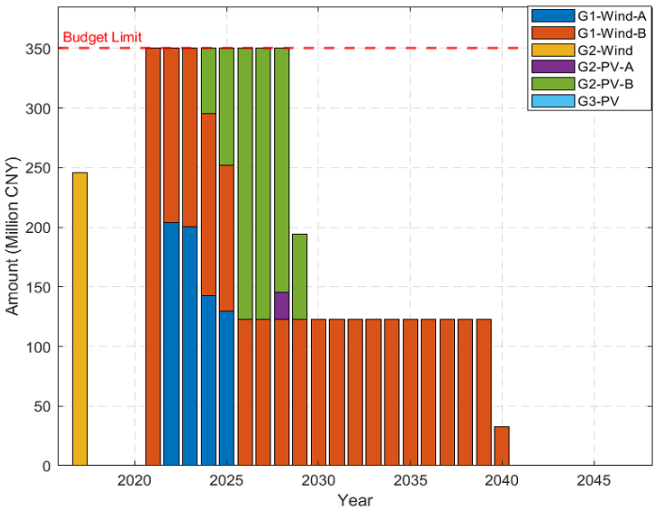


Figure 16: Dynamic optimal allocation of subsidies under the carbon price depreciation scenario

Although recent empirical trends generally point toward carbon price appreciation, the depreciation scenario provides a useful stress test for green finance policy design. The results quantify the fiscal cost of carbon market downside risk and show that subsidy capacity must be adjustable when market-based carbon revenues weaken. More broadly, the dynamic analysis demonstrates that the proposed framework can recalibrate policy instruments in response to carbon price movements, thereby preserving instrument solvency under both favorable and adverse market conditions. A further sensitivity analysis of carbon price fluctuations is provided in Appendix A.1.

7. Economic Mechanisms and Market Implications

7.1. Policy optimization and financial market transmission

This study develops a unified framework for analyzing how government policy can be designed to internalize environmental externalities and transmit into green bond valuation. Our findings show that carbon credits and subsidies serve distinct but complementary roles. Carbon credits operate as the first-best market-based channel by converting verified emission reductions into project cash flows. When carbon-market revenues are insufficient to restore financial viability, fiscal subsidies provide a targeted backstop. This hierarchical structure preserves the role of market incentives, while reserving the public funds to projects whose financing gaps cannot be closed through CCER revenues alone.

The timing of intervention is central to the efficiency of this mechanism. In the absence of binding constraints, the optimization problem favors front-loaded allocations that minimize the nominal policy resource commitment. Under binding credit or fiscal caps, however, the allocation shifts toward maximizing the marginal effectiveness of scarce policy resources. Tighter constraints force support into less efficient periods and increase the nominal resource requirement needed to achieve the same viability target. The dynamic analysis further shows that this allocation must adjust to carbon price movements. Carbon price appreciation reduces the need for fiscal subsidies by increasing

the value of CCER revenues, whereas carbon price depreciation raises the fiscal burden and requires greater budget flexibility.

These optimized interventions transmit to the financial market by changing the risk-return profile of the underlying green assets. Policy support does not simply fill a funding gap, it reduces downside viability risk and moves the embedded option from a deep OTM state toward an ATM state by repairing project cash flows. Once this cash-flow shortfall is resolved, project volatility becomes a source of priced upside potential rather than merely a source of risk. This mechanism provides a structural foundation for the environmental premium. Policy intervention converts otherwise unpriced environmental benefits into cashflow relevant values that can be capitalized into green bond prices. In this sense, the model proposed in this paper separates the conventional financial component of green bond value from the policy-induced environmental component, and quantifies the latter as the market value of externality internalization under credible policy support. This distinction clarifies how well-designed intervention can mobilize private capital while preserving fiscal discipline, because public support is targeted toward restoring project viability rather than generating excess compensation.

7.2. Implications for policymakers, investors, and issuers

The framework has several implications for the design and pricing of green finance instruments. For policymakers, the results support a market-first intervention hierarchy. Carbon-credit mechanisms should be prioritized when verified emission reductions can generate sufficient project cash flows, while subsidies should be reserved for residual financing gaps. At the same time, fiscal rules must remain sufficiently flexible to respond to carbon-market downturns. A rigid subsidy cap may be cost-efficient under baseline conditions but insufficient when carbon prices decline and CCER revenues weaken. Effective policy design should therefore combine fiscal discipline with adaptive capacity.

For investors, the option-based pricing model provides a way to evaluate the latent value of green bonds under uncertainty. Traditional static NPV measures can understate the value of green assets once policy support repairs the downside cash-flow gap and activates upside optionality. The embedded green option captures this incremental value and provides a structural basis for interpreting the greenium. Investors should therefore assess green bonds not only by their contractual cash flows and issuer credit risk, but also by the credibility of policy support, the verifiability of environmental cash flows, and the extent to which these cash flows are pledgeable to bondholders.

For issuers, the results suggest that green bond design should better align debt service obligations with the timing of policy-induced cash flows. Performance-linked structures, step-up or step-down coupon features, and project-pool designs can help connect financing terms to verified environmental outcomes and policy support. Such structures may reduce the mismatch between upfront investment costs and delayed environmental revenues, thereby improving investor confidence and lowering financing costs. By quantifying the embedded green option, issuers can also communicate the economic value of policy-supported environmental performance more clearly to the market.

In summary, the proposed framework shows how policy design and asset pricing interact in green finance. By linking optimized intervention to project cash-flow repair and embedded option

valuation, the model provides an endogenous transmission mechanism from environmental policy to green bond pricing. This mechanism helps explain when green assets become financially investable, how the greenium emerges, and why policy boundaries must be aligned with the structure of financial instruments to mobilize private capital efficiently.

8. Conclusion

This paper develops a two-stage framework that links environmental policy optimization to green bond pricing. The first stage determines the cost-efficient mix of carbon credits and fiscal subsidies needed to restore the financial viability of green projects. By embedding a market-first hierarchy, the framework prioritizes CCER revenues when verified emission reductions can support project cash flows and uses subsidies only as a residual fiscal backstop. The results show that coordinated carbon-credit and subsidy interventions can maintain project solvency under credit-supply and fiscal-budget constraints while limiting unnecessary public resource use. The robustness and dynamic analyses further show that the optimal policy mix is stable under alternative viability targets and intervention boundaries, but must be recalibrated when carbon prices change. In particular, carbon price appreciation reduces fiscal reliance by increasing the value of CCER revenues, whereas carbon price depreciation raises the subsidy capacity required to preserve solvency.

The second stage translates the optimized policy outcomes into green bond pricing implications. By incorporating policy-adjusted project cash flows into an option-pricing framework, the model decomposes green bond value into a conventional fixed-income component and an embedded green option. Policy intervention improves the risk-return profile of green assets by moving the embedded option from a distressed OTM state toward a repaired ATM state, thereby allowing project volatility to become a source of priced upside potential rather than merely downside risk. The framework proposed in this paper provides a structural interpretation of the greenium by showing how policy support internalizes environmental value into project cash flows and enables the market to capitalize the resulting option-like payoff. Our findings highlight the importance of aligning policy intervention boundaries with the structure of financial instruments. Effective green finance policy should combine market-based incentives, disciplined fiscal support, and risk-sensitive pricing mechanisms to mobilize private capital for the low-carbon transition. Future research may extend the framework by allowing policy preference parameters to evolve dynamically and by incorporating broader welfare or emissions-reduction objectives beyond financial break-even.

Acknowledgments

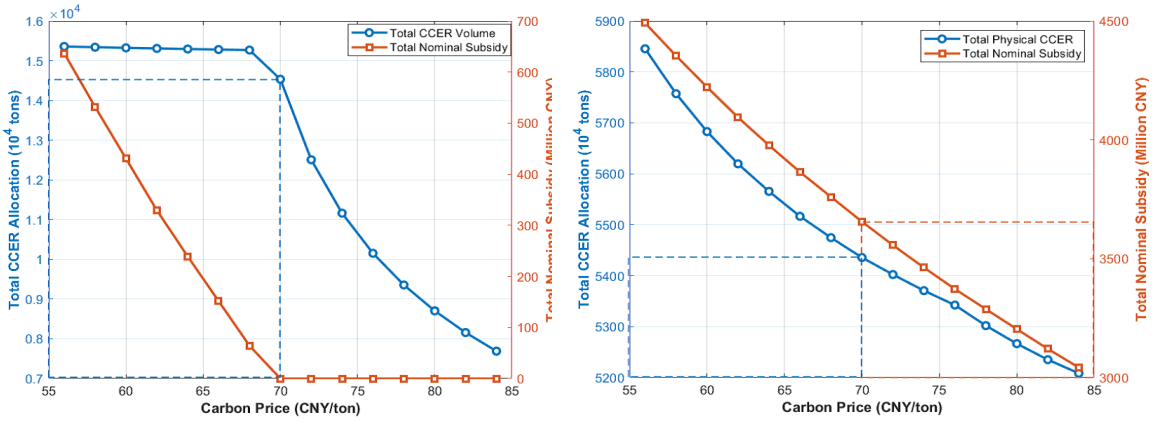
The authors would like to thank the editors and the anonymous reviewers for their valuable comments and suggestions which have helped immensely in improving the quality of this paper. This work was supported by National Social Science Fund of China (24BJY216), the Singapore Ministry of Education (MOE) Academic Research Fund (AcRF) Tier 3 (MOE-MOET32022-0006), and the Singapore Ministry of Education (MOE) Academic Research (AcRF) Fund Tier 1 (RG137/25).

Appendix A. Sensitivity analysis

Appendix A.1. Sensitivity analysis of carbon trading price p_l^c

Theoretically, carbon credits mitigate the deficit by internalizing environmental externalities. The present value of each credit is defined as $p_l^c/(1+r)^l$, where a higher p_l^c enhances the marginal economic value of each unit. However, when baseline emissions exceed its reductions ($v_{i,j,l} > a_{i,j,l}$), a rising p_l^c would increase emission costs by $-p_l^c(v_{i,j,l} - a_{i,j,l})$, creating two opposing effects: the enhanced revenue per credit versus an expanding financial shortfall. Since our case is a renewable energy project, we adopt a counterfactual baseline where $v_{i,j,l} = a_{i,j,l}$, effectively offsetting the cost term. Consequently, the marginal efficiency of carbon credits improves monotonically with p_l^c, therefore the government support required to achieve solvency decreases. To test this mechanism, we examine carbon price fluctuations within a $\pm 20\%$ range.

As shown in Fig. A.1, a clear inverse relationship exists between the carbon price p_l^c and government support required, consistent with our expectations. Specifically, as shown in Fig. A.1(a), under the global constraint $G_l = 5.5$ million tons, we identify $p_l^c = 70$ CNY/ton as the threshold where subsidies are phased out. When $p_l^c < 70$, carbon credits alone are insufficient to bridge the deficit, triggering supplementary subsidies. Although the model exhausts the G_l cap in high-demand years, in marginal years where credit demand falls below the cap, higher carbon prices reduce the required physical emission reduction per unit of output. As a result, the total credit volume still exhibits a slight decline as p_l^c increases. Conversely, when $p_l^c > 70$, the carbon credits constraint becomes non-binding, eliminating the need for subsidies and allowing the required credit volume to decrease as the revenue generated per unit of credit increases.



(a) with constraint $G_l = 5.50$ million tons (b) with constraint ${v^b}_{i,j,l} \leq a_{i,j,l}$

Figure A.1: Sensitivity of optimal intervention to carbon price fluctuations

Note: The two panels compare different constraint regimes: panel (a) focuses on the global cap G_l, where subsidies act as a safety net when market-based credits reach their limit. In contrast, panel (b) applies the tighter ${v^b}_{i,j,l} \leq a_{i,j,l}$ constraint, requiring persistent subsidy intervention to compensate for the individual projects' physical credit caps. For the global cap G_l, the baseline value $B_l = 310$ million CNY is maintained, whereas for the tighter constraint ${v^b}_{i,j,l} \leq a_{i,j,l}$, the cap is relaxed to $B_l = 350$ million CNY.

Furthermore, under the stricter physical constraint ${v^b}_{i,j,l} \leq a_{i,j,l}$, as shown in Fig. A.1(b), both subsidies and credits decrease simultaneously as p_l^c rises. In this scenario, since credits are strictly bounded by physical limits, carbon revenue alone cannot fully offset the deficit, requiring subsidy supplementation. However, as the marginal value of carbon credits increases, enhanced revenue efficiency effectively narrows the funding gap. This leads to a prioritized phase-out of subsidies followed by a reduction in credit demand, demonstrating the strong compensatory power of a robust carbon price in ensuring project viability.

While subsidies act as an effective catalysts for green investment, over-reliance can impose fiscal strains and long-term sustainability risks. This analysis demonstrates that governments can leverage the carbon market as a strategic substitute for direct fiscal expenditure. By fostering a robust carbon price, policymakers can achieve optimal resource allocation under stringent budgetary constraints.

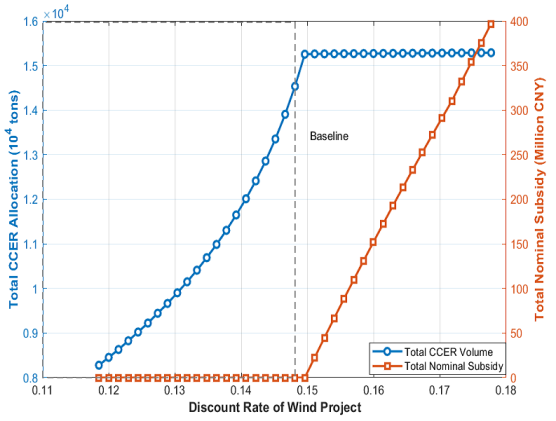
Appendix A.2. Sensitivity analysis of discount rate r

We now evaluate the sensitivity of optimal policy allocations to the discount rate (r), which reflects the opportunity cost of capital and the risk premium required by investors. A higher discount rate implies a stronger preference for immediate returns over delayed cash flows. It also generates two opposing effects on intervention requirements: it reduces the present value of the project's deficit, thereby lowering required support, while simultaneously diminishing the present value of future credit allocations and subsidies, weakening their per-unit compensatory effectiveness.

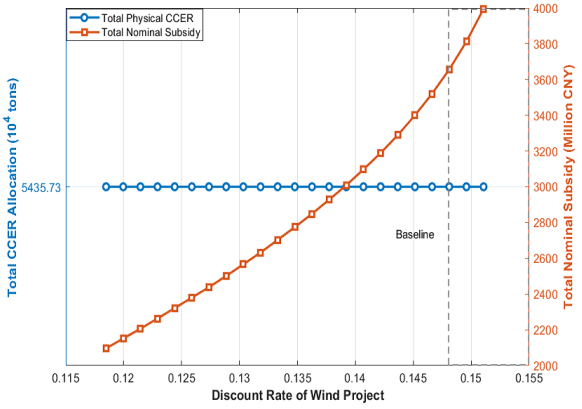
For this analysis, we vary the baseline discount rates for both wind and PV projects by $\pm 20\%$, with the resulting trajectories depicted in Fig. A.2. The results show that the latter effect dominates: the intertemporal value erosion of the policy instruments outweighs the reduction in project deficits. Consequently, the model must allocate a larger nominal volume of credits and subsidies to maintain the break-even condition, resulting in a positive correlation between the discount rate and intervention volume until binding constraints are reached.

The behavioral patterns of policy intervention vary distinctly across the two constraint regimes, directly corresponding to the solutions reported in Column 8 of Table 2, Column 2 and 3 of Table 3. As shown in panel (a) and panel (c) of Fig A.2, with the annual global cap $G_l = 5.50$ million tons, the total CCER volume eventually reaches a plateau. Once the intertemporal demand for CCERs exceeds the annual regulatory cap, fiscal subsidies are activated to bridge the remaining financial gap and increase alongside the discount rate. In contrast, for wind projects under the individual physical limit ${v^b}_{i,j,l} \leq a_{i,j,l}$, credit allocations reach their physical caps much more rapidly. As shown in panel (b), the credit allocation for wind projects flattens at 54.36 million tons (matching Column 2 of Table 3), indicating that these projects quickly exhaust their credit capacities and leave all residual deficits to be exclusively absorbed by subsidies.

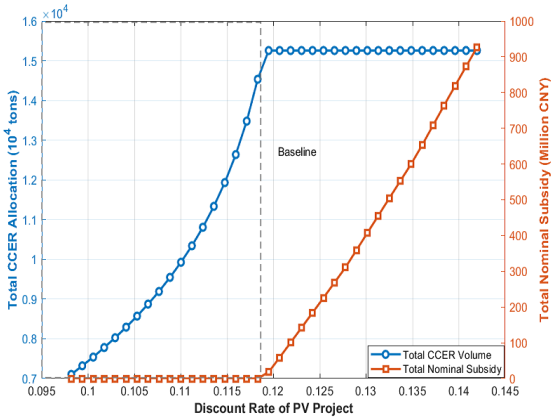
Panel (d) in Fig. A.2 demonstrates a more complex, three-phase evolution of policy intervention as r increases. In Phase I, as r begins to rise, the expanding deficit of PV projects is absorbed solely through increased CCER allocations, while subsidies remain constant at the baseline level required to support wind projects, since the PV projects still possess sufficient credit capacity. As r continues to rise, the system enters Phase II characterized by a simultaneous increase in both CCERs and subsidies. This dual expansion is driven by portfolio heterogeneity: certain PV projects hit their



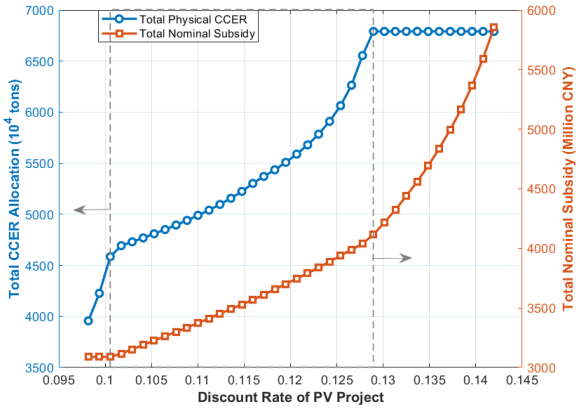
(a) with constraint $G_l = 5.5$ million tons



(b) with constraint $v^b{}_{i,j,l} \leq a_{i,j,l}$



(c) with constraint $G_l = 5.5$ million tons



(d) with constraint $v^b{}_{i,j,l} \leq a_{i,j,l}$

Figure A.2: Sensitivity of optimal intervention allocation to the discount rate (r)

Note: The tighter constraint $v^b{}_{i,j,l} \leq a_{i,j,l}$ requires persistent subsidy. Therefore, for constraint G_l=5.5 million tons, $B_l = 310$ million CNY is maintained, for constraint $v^b{}_{i,j,l} \leq a_{i,j,l}$, the cap is relaxed to $B_l = 350$ million CNY. While excessively high discount rates lead to no feasible solution for the optimization model, therefore, the horizontal axes in panel (b) is truncated to display only the valid discount rate intervals where the optimal solution exists. Minor discrepancies may arise due to rounding at different stages of computation. All values are reported to two decimal places for presentation purposes.

individual physical limits ($v^b_{i,j,l} = a_{i,j,l}$) and trigger subsidy support, while others still have remaining credit capacity to expand. Finally, in Phase III, the total CCER allocation reaches its physical cap of 67.91 million tons (matching Column 3 of Table 3), completely exhausting the portfolio's market-based capacity. Consequently, any further increase in deficit must be met entirely by fiscal subsidies, resulting in a plateaued credit curve and a steeply rising subsidy trajectory.

The observed positive correlation between the discount rate and intervention intensity is fundamentally driven by a duration mismatch between project deficits and policy interventions. Since capital expenditures are largely upfront while policy benefits are deferred, high interest rates erode policy effectiveness faster than they reduce project-level financial stress. Therefore, for policymakers, it is inefficient to rely solely on future-dated credits in high-interest-rate or high-risk environments due to the severe intertemporal value erosion. During periods of monetary tightening or heightened market risk, governments should implement front-loaded support or dynamic subsidy injections to counteract the adverse discounting effect and prevent market failure.

Appendix A.3. Sensitivity analysis of the volatility σ

The crucial link between policy optimization and bond pricing lies in how government intervention unlocks the value of volatility. Volatility σ quantifies the uncertainty in a project's future revenue, a higher σ implies a wider distribution of potential outcomes. To assess its impact, we conduct a sensitivity analysis by varying σ by $\pm 20\%$ from its baseline based on *Scenario 4: Synergistic optimization*. Since policy intervention ensures a zero-NPV state, the underlying valuation inputs remain consistent across different schemes, making this pricing analysis broadly applicable.

The sensitivity analysis in Fig. A.3 reveals two primary insights. First, the valuation of all embedded green options is positively correlated with volatility, aligning with option pricing theory where higher volatility increases potential upside gains while downside risk is capped. Second, the sensitivity to volatility varies across the three bonds due to scale and leverage effects. As summarized in Table A.1, although 22 Jidian G1 has the highest absolute price fluctuation, it exhibits the lowest relative sensitivity to volatility. In contrast, 22 Jidian G2 and 23 Jidian GCKV01 demonstrates a greater dependence on future environmental uncertainty to unlock their valuation potential.

Table A.1: Sensitivity of green option value to fluctuations in asset volatility (σ)

Bond Name	Green Option C (million CNY)			Abs. Range	Percentage Change		
	$\sigma_{low}(-20\%)$	σ_{base}=10.69%	$\sigma_{high}(+20\%)$	(million CNY)	ΔC_{low}	ΔC_{high}	Mean Sens.
22 Jidian G1	91.15	113.85	136.47	45.32	−19.94	+19.87	19.91
22 Jidian G2	69.71	87.09	104.45	34.74	−19.96	+19.93	19.95
23 Jidian GCKV01	63.39	79.19	94.97	31.58	−19.95	+19.93	19.94

Note: $\sigma_{base} = 10.69\%$. Abs. Range is defined as the absolute difference between C_{high} and C_{low}. Percentage changes (ΔC) are relative to the baseline C_{base}. Mean Sensitivity represents the average absolute percentage deviation.

Then, we evaluate how the implied yield (y_g) and greenium respond to sudden shifts in market uncertainty. Using 22 Jidian G1 as an example, we conduct a stress test by exposing the valuation framework to three volatility regimes: a low-volatility scenario representing a stable energy market, the baseline scenario, and a high-volatility shock reflecting extreme market fluctuations. The stress test results in Table A.2 further confirm that the greenium increases with higher systemic risk.

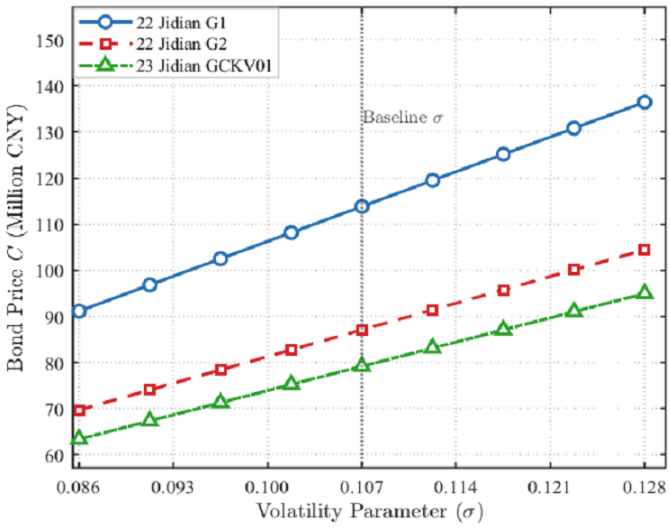


Figure A.3: Sensitivity of green option pricing to asset volatility σ

Table A.2: Valuation components and implied yield for 22 Jidian G1 under different σ

Scenario	Principal (million CNY)	Fixed Value P_{fixed} (million CNY)	Green Option C (million CNY)	Total Value P_{GB} (million CNY)	Implied Yield	Greenium (bps)	Greenium Gaps (bps)
σ_{low}=8.55%	1000	1023.69	91.15	1114.84	1.41%	187.45	239.45
σ_{base}=10.69%	1000	1023.69	113.85	1137.54	0.97%	231.13	283.13
σ_{high}=12.83%	1000	1023.69	136.47	1160.16	0.54%	273.57	325.57

Specifically, in the low-volatility scenario ($\sigma_{low} = 8.55\%$), the green option value is $C = 91.15$ million CNY. Although lower market volatility limits the upside potential of environmental assets, the bond still exhibits a theoretical greenium of 187.45 bps, indicating the presence of a policy-induced valuation floor. In the high-volatility shock scenario ($\sigma_{high} = 12.83\%$), the option value increases to $C = 136.47$ million CNY, representing a 19.87% increase relative to the baseline. Consequently, the theoretical greenium expands to 273.57 bps.

These results demonstrate that, in the valuation of green bonds with embedded options, asset volatility is a key determinant of the value of the embedded environmental option. The synergistic policy framework effectively transforms green assets into volatility-sensitive instruments, where higher market uncertainty amplifies the economic value of embedded green options. In this context, volatility is no longer merely a source of risk; instead, it serves as a proxy for the potential release of policy-driven value. Consequently, volatility should be explicitly considered in pricing to ensure that green instruments are valued at their intrinsic economic worth.

Appendix A.4. Sensitivity analysis of the policy preference ratio $\rho = \lambda_2/\lambda_1$

To examine the robustness of the proposed market-first intervention hierarchy, we conduct a sensitivity analysis on the policy instrument weights. The ratio $\rho = \lambda_2/\lambda_1$ measures the regulator's relative preference for market-based carbon-credit support over direct fiscal subsidies. A larger value of ρ implies that subsidies receive a higher penalty within the objective function and are therefore used only when market-based instruments become insufficient. Throughout this experiment, we normalize $\lambda_1 = 1$ and vary $\lambda_2 \in [0, 200]$, with the resulting allocation paths shown in Fig. A.4.

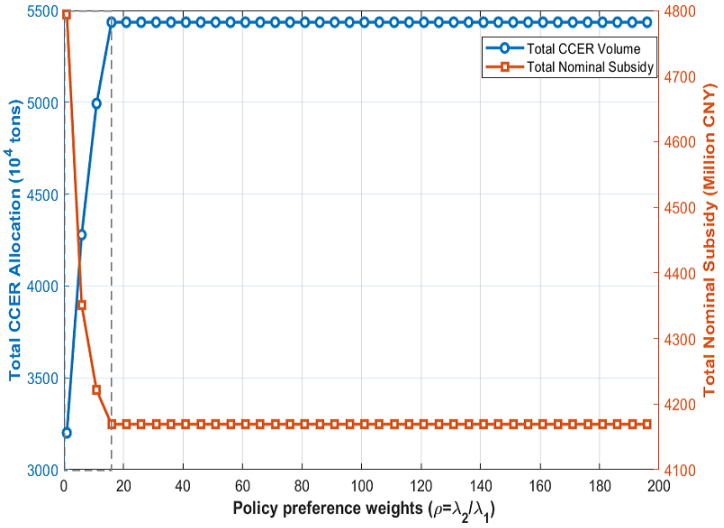


Figure A.4: Sensitivity of optimal intervention to policy preference ratio ($\rho = \lambda_2/\lambda_1$)

The results reveal a clear two-stage transition in the optimal intervention strategy. When ρ is relatively small, subsidies receive only a mild penalty and therefore constitute the dominant policy instrument. As ρ increases, fiscal support becomes progressively more costly, inducing the optimizer to substitute carbon credits for subsidies whenever feasible. Consequently, nominal subsidy expenditure declines rapidly, while CCER allocation increases correspondingly. Once ρ exceeds the critical threshold of $\rho = 16$, the total CCER allocation reaches its physical limit imposed by the constraint ${v^b}_{i,j,l} \leq a_{i,j,l}$, consistent with Column 2 of Table 3. Beyond this point, further increases in ρ no longer affect the allocation pattern, as the system enters a steady regime governed by the binding physical constraint rather than the penalty parameter. The CCER allocation remains flat because the available emission reductions are fully exhausted, while a minimum fiscal subsidy requirement remains binding to ensure project viability.

This sensitivity analysis demonstrates that the market-first intervention hierarchy is not an artifact of a particular parameter calibration but rather a structural implication of the optimization framework. Once the regulator assigns sufficiently greater weight to subsidies than to carbon credits, the model consistently exhausts market-based environmental assets before activating fiscal backstops. The transition from a substitution region to a plateau further reflects the finite physical capacity of carbon-credit generation and provides additional justification for integrating market mechanisms with complementary fiscal support.

Appendix A.5. Sensitivity analysis of the objective trade-off weight μ

We next examine the sensitivity of the optimization results to the objective trade-off weight μ. As defined in Eq. (7), μ governs the trade-off between minimizing the discounted policy-preference cost and minimizing the nominal policy resource commitment. A larger value of μ places greater emphasis on discounted costs, whereas a smaller value emphasizes the total nominal amount of policy resources deployed. Figure A.5 illustrates the resulting allocation trajectories.

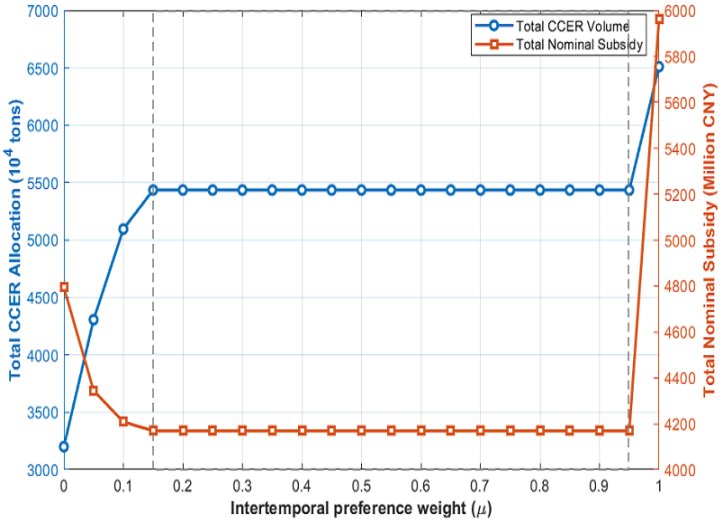


Figure A.5: Sensitivity of optimal intervention to the objective trade-off weight μ

When $\mu \to 0$, the optimization is driven primarily by nominal policy resource commitments rather than their discounted values. Since this component does not incorporate policy preference weights (λ_1, λ_2), subsidies become relatively more attractive and are used more extensively, while CCER allocation remains comparatively low. As μ increases, the discounted policy-preference cost becomes more influential, inducing the optimizer to increasingly substitute market-based CCER support for fiscal subsidies. Consequently, subsidy expenditure declines while CCER allocation rises, until the available carbon-credit capacity is exhausted. This regime is consistent with the range of $\rho < 16$ shown in Fig. A.4.

Across a broad intermediate range of $\mu \in [0.15, 0.95]$, both policy instruments stabilize at nearly constant levels. This stable region corresponds closely to the plateau observed for $\rho > 16$ in Fig. A.4, where further changes in the policy preference ratio no longer affect the optimal allocation. Together, these results indicate that the optimizer has identified an allocation that balances discounted policy costs and nominal resource commitments while satisfying both the project viability constraint and the limits on policy support. The resulting allocation is therefore robust and relatively insensitive to moderate changes in the trade-off parameter μ.

When $\mu \to 1$, the optimization objective is dominated almost entirely by discounted policy costs, with little consideration given to nominal resource commitment. Under this specification, substantially larger nominal allocations may be required to achieve equivalent discounted compensation, particularly for interventions occurring in later years. Consistent with Fig. 2(a) in Section 3.2.2, both CCER allocation and subsidy expenditure increase sharply. Moreover, optimization driven solely by discounted costs becomes more susceptible to multiple timing patterns that yield similar present-value outcomes. This observation further justifies the inclusion of the nominal policy resource commitment term, which helps distinguish economically meaningful allocation paths and produces more stable optimization results.

Accordingly, the baseline analysis adopts $\mu = 0.5$ as a balanced benchmark, assigning equal

importance to discounted policy costs and nominal policy resource commitment. The sensitivity results demonstrate that the main conclusions remain stable across a wide range of μ, indicating that the proposed intervention mechanism is not driven by a particular choice of weighting parameter.

Appendix B. The information on projects supported by bonds

See Table B.3.

Table B.3: Basic information and financing parameters of the supported green projects

Bond Group	Project Name	Start Year	Duration (Years)	Type	Est. Generation (MWh)	Reductions (10^4 tons)	Total Capex (million CNY)	Bond Share (million CNY)	Ratio (ϕ)
22 Jidian G1	G1-Wind-A	2022	20	Wind	819,300.00	67.75	2265.00	647.50	28.59%
	G1-Wind-B	2021	20	Wind	545,929.20	39.05	2377.85	300.00	12.62%
22 Jidian G2	G2-Wind	2017	20	Wind	98,310.60	7.66	456.37	300.00	65.74%
	G2-PV-A	2023	25	PV	80,098.00	5.54	200.00	50.00	25.00%
	G2-PV-B	2022	25	PV	1,148,445.56	89.25	3669.51	350.00	9.54%
23 Jidian GCKV01	G3-PV	2022	25	PV	1,062,440.00	84.35	2742.00	478.00	17.43%

Note: Total Capex (million CNY) is the total capital expenditure for the project, Est. Generation(MWh) is the estimated annual electricity output over the project's operational life, as projected in feasibility studies or certified assessments, sourced from official bond prospectuses and third-party certification reports.

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